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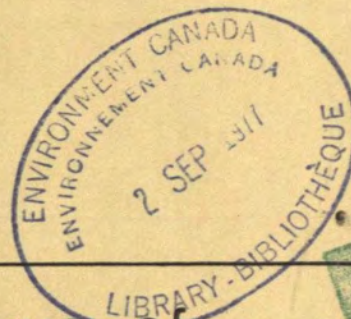
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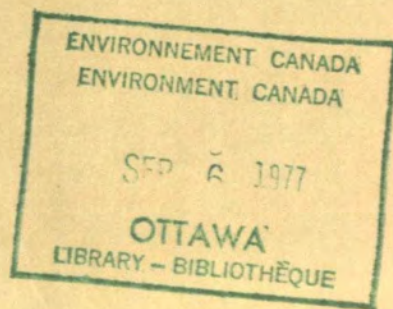
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and Environment
Canada

Pêches
et Environnement
Canada



Preliminary Catalogue of Salmon Streams and Spawning Escapements of Statistical Areas 19 & 20 (Victoria - Sooke)

D.E. Marshall
R.F. Brown
V.D. Chahley
D.G. Demontier



L Data Record Series
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L Pacific Region
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Preliminary Catalogue of Salmon Streams and Spawning Escapements of Statistical Areas 19 & 20 (Victoria – Sooke)

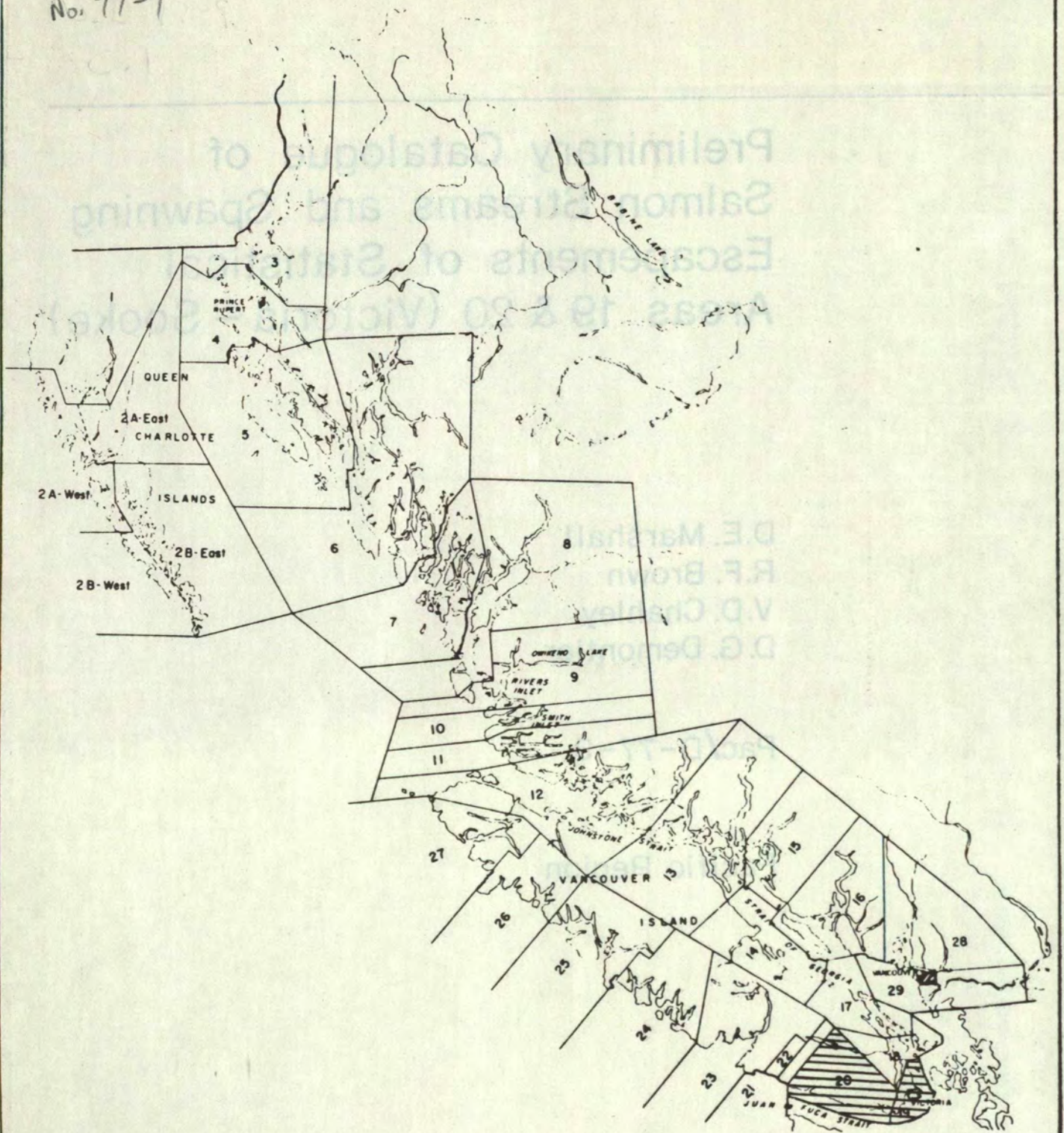
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D.G. Demontier

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Pacific Region

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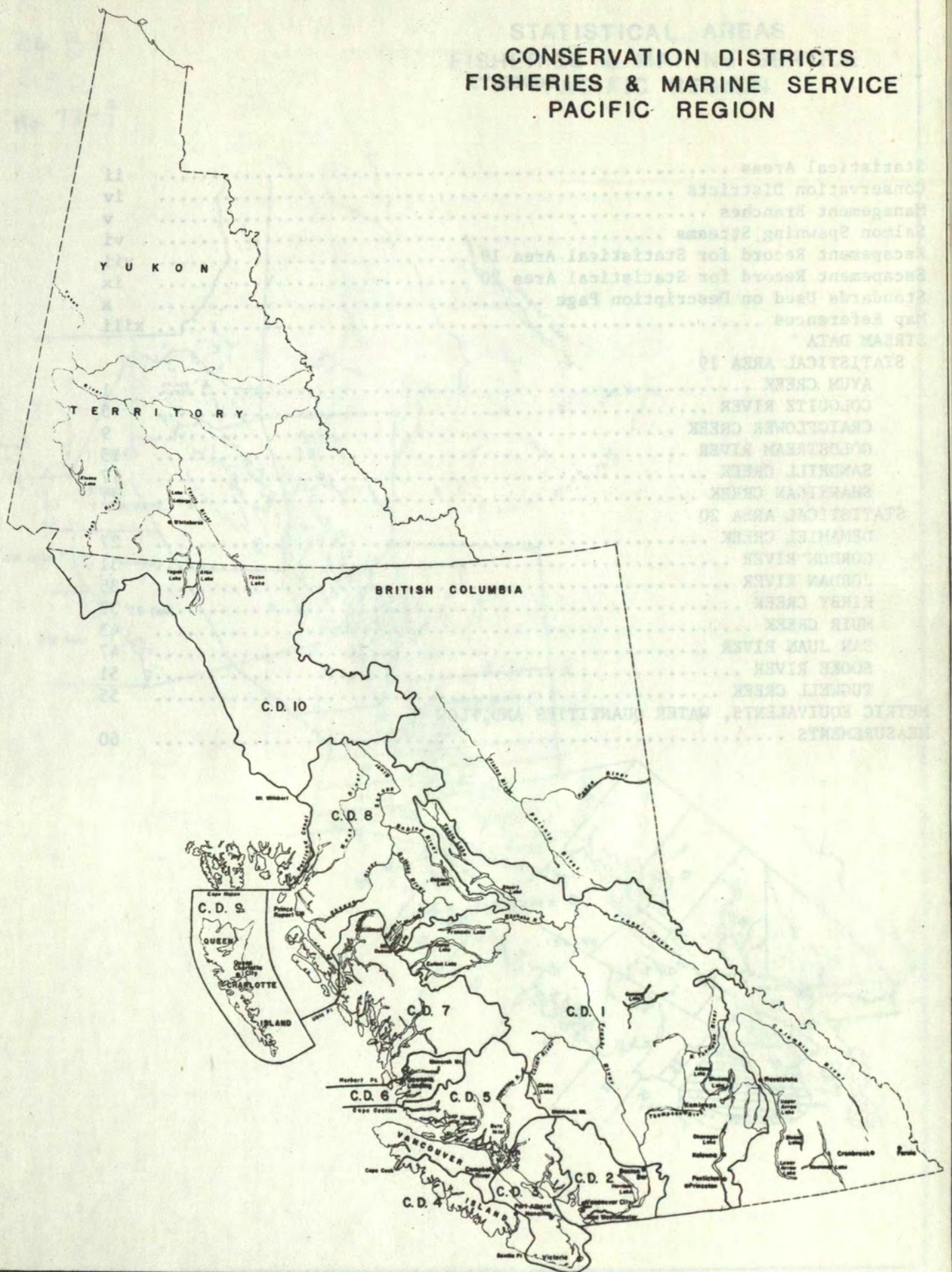
STATISTICAL AREAS FISHERIES & MARINE SERVICE PACIFIC REGION



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STATISTICAL AREAS CONSERVATION DISTRICTS FISHERIES & MARINE SERVICE PACIFIC REGION

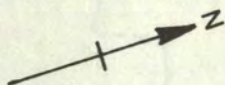


MANAGEMENT BRANCHES
FISHERIES & MARINE SERVICE
PACIFIC REGION



SALMON SPAWNING STREAMS
STATISTICAL AREAS 19 & 20

0 10
miles



Port
Renfrew

Juan
de
Fuca
Strait

Statistical
Area 20

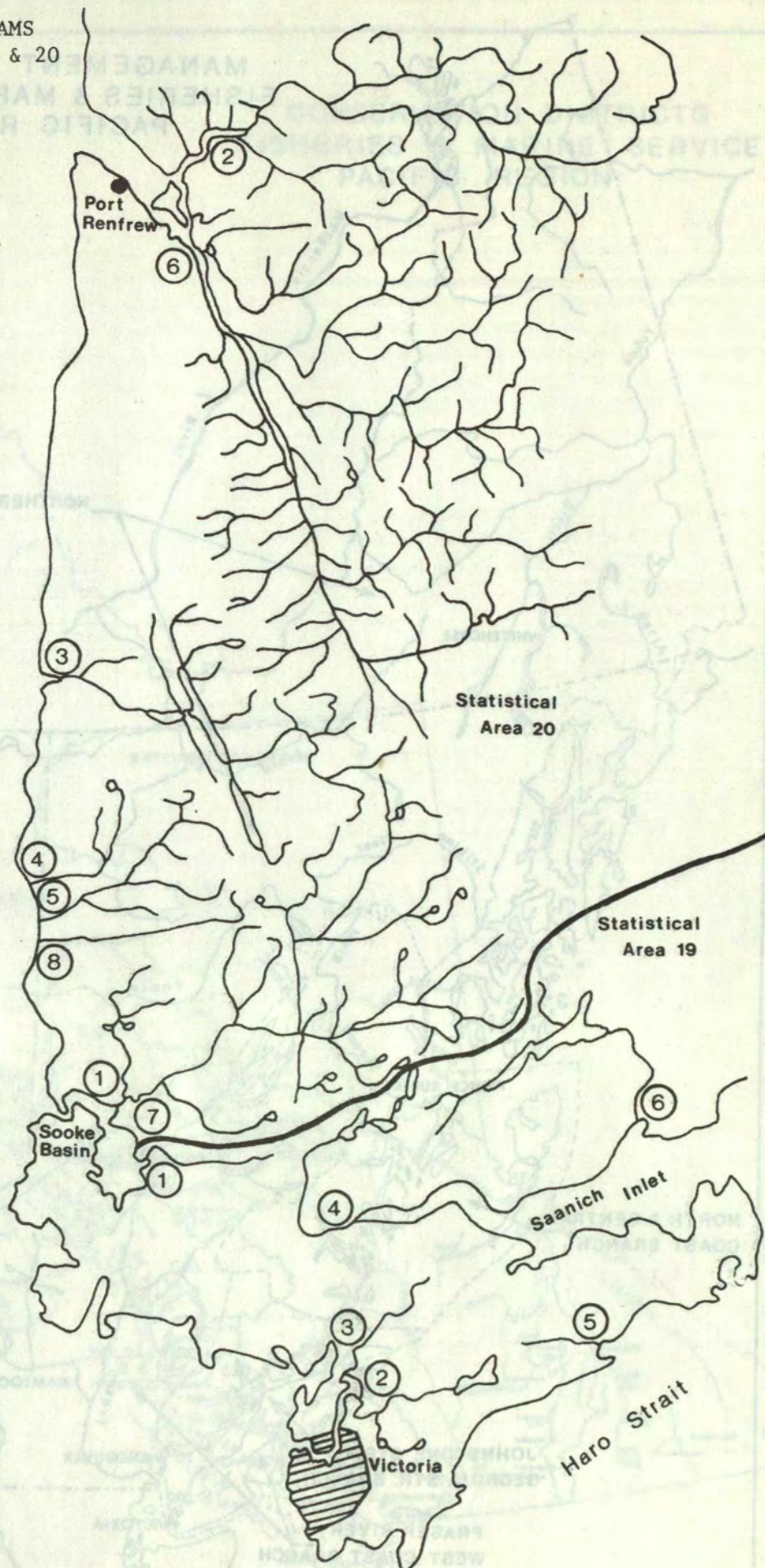
Statistical
Area 19

Statistical Area 19

1. Ayum Creek
2. Colquitz River
3. Craigflower Creek
4. Goldstream River
5. Sandhill Creek
6. Shawnigan Creek

Statistical Area 20

1. DeMamiel Creek
2. Gordon River
3. Jordan River
4. Kirby Creek
5. Muir Creek
6. San Juan River
7. Sooke River
8. Tugwell Creek



ESCAPEMENT RECORD FOR STATISTICAL AREA 19

YEAR	SOCKEYE	CHINOGK	COHO	CHUM	PINK	STEELHEAD
1947		2	3050	23100		
48		5	2500	42500		
49			1100	38500		
50			1925	38500		
51			2050	18500		
52			2325	35200		
53			1025	11000		
54			500	26000		
55			125	10500		
56			950	7150		
57			150	9750		
58			475	12500		
59			475	4100		
60			600	2300		
61			700	7550		
62		25	250	8450		
63		2	600	3550		
64			2300	5650		
65		25	400	12825		
66		25	1165	10025		
67		40	420	7775		
68			717	10700		
69		23	643	8950		
70		25	2250	11075		
71		25	1725	7000		
72		25	325	7000		
73		25	272	11000		
74		25	688	9000		
75		75	400	1900		
76		25	250	3900		
77						
78						
79						
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81						
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83						
84						
85						
Time						
Arr.						
Start						
Peak						
End						

REMARKS

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ESCAPEMENT RECORD FOR STATISTICAL AREA 20

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947		400	11150	23400		
1948		1150	32750	33450	25	
1949		600	7000	50025	75	
1950		2250	12100	49700	75	
1951		1150	25250	57675	75	
1952		1150	12275	18275	10500	
1953		600	3850	60075		
1954		675	7150	52200	3000	
1955		300	4150	51675	75	
1956		50	15525	22950	3000	
1957		250	8950	23850		
1958		600	20800	75650	1900	
1959		525	9275	29400		
1960	75	2050	23000	10075	2650	
1961	75	675	37050	9200		
1962	76	1350	22575	13025	8650	
1963	25	2650	14950	12400		
1964	25	2450	80525	52675	3500	
1965	1500	1350	26350	21600		
1966	1500	2325	24100	16225	5000	
1967	1500	2650	37350	44475	1150	
1968	1500	4650	19400	75850	500	
1969	200	2100	16150	51200		
1970	3500	9750	82100	59075	1500	
1971	25	1900	9826	22625		
1972	1500	7750	20225	93725		
1973		7400	25383	111037		
1974	75	1152	10825	19675	75	
1975	77	625	2625	7650		
1976	83	105	1350	5825	75	
1977						
1978						
1979						
1980						
1981						
1982						
1983						
1984						
1985						
Time						
Arr.						
Start						
Peak						
End						

REMARKS

STANDARDS USED ON STREAM DATA PAGE

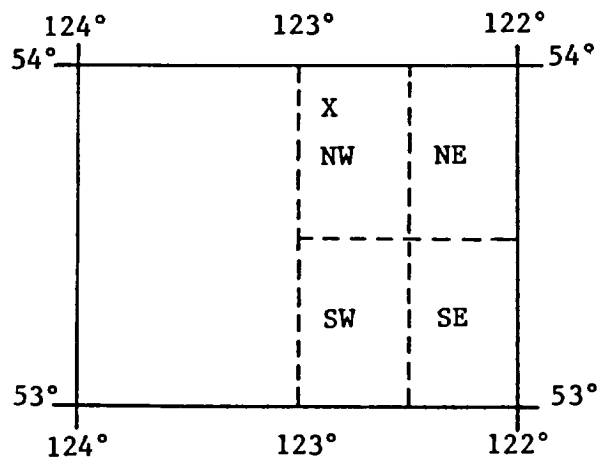
Name of Stream: Name as given in Gazetteer of Canada, British Columbia edition; local names are added in lower case type.

Conservation District: As defined by the Conservation and Protection Service (April 1965).

Statistical Area: As defined by Department of Fisheries Statistical Map (June 1957).

Location and Position: Defined by quadrant indexing. Each geographical quadrilateral of the earth's surface of 1 degree in extent in latitude and longitude is divided into the SE, SW, NE and NW quarters. The south-east corner of each quadrilateral gives the initial point for the figure of reference (Gazetteer of Canada).

EXAMPLE "X"
53° 122° NW



Length: The portion of the stream utilized by spawning salmon.

Width: Average width, estimated to the nearest foot for the described length.

Drainage: Area in square miles of the entire drainage basin feeding the stream.

Composition:

Bedrock	bedrock
Boulder	>256 mm (>10")
Coarse	50.9 - 256 mm (2 - 10")
Fine	3.37 - 50.8 mm (1/8 - 2")
Sand & Silt	<3.37 mm
Unclassified	where bottom cannot be observed, eg. log jam, pools, water colour, etc.

Gradient: Average vertical drop per thousand linear feet.

Wetted Area: Number of square yards of stream bed under water at average flows within the described length.

Spawning Area: Estimated square yardage of stream bed suitable for salmon spawning within the described length.

Discharge: Mean annual discharge. Maximum and minimum values are either daily means or instantaneous discharges. The latter are identified by (Inst.). Discharge data is taken from "Historical Stream Flow Summary", British Columbia, Water Survey of Canada.

Temperature: As described.

Barriers and Points of Difficult Ascent: Complete and partial barriers to salmon and their distance from the stream mouth. Species likely to be affected may be listed. Both natural and man-made obstructions are defined.

Spawning Distribution: Distribution is indicated by brief comments opposite the species.

Potential of Inaccessible Portion of Stream: As described.

General Remarks: Emphasizes features of stream and spawning populations. Also includes industrial activity, routes of accessibility, etc. The comments with dates following them are taken from "Annual Reports of Salmon Stream & Spawning Grounds" (B.C. 16's).

Escapement Record: The escapement represents the mid point of the coded range of escapement for each species. For example: 5000-10000 would be entered as 7500. Where absolute numbers are provided by Fisheries Personnel, these numbers are entered. N/O means the stream was inspected and no fish were observed; UNK means evidence of fish present but no estimates were made; NO RECORDS means no escapement records for the applicable years could be found in the escapement files.

Timing: Arr. - arrival in stream

Spawning:

E - early (first 10 days of month)

M - middle (middle 10 days of month)

L - late (last 10 days of month)

Distance references are from mouth of stream, unless otherwise stated.

R
L
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L
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A
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C
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C
C
C
F
L

MAP REFERENCES

Roads:

hard surface, all weather	more than 2 lanes
hard surface, all weather	2 lanes ^{Route No.} less than 2
loose surface, all weather	2 lanes wide or more
" less than 2 lanes	all weather dry weather

Cart Track, Trail --- Cart Track Trail

Railways:

normal gauge, multiple track	Station
normal gauge, single track	Stop
abandoned, or under construction	Siding
narrow gauge, single track	
Bridge, underpass or overpass	
Tunnel	

Boundary, International	
" Province	
" County or District	
" Township or Parish	
" City or Town	
" Reservation, Indian, Military, etc	
Power Transmission Line	
Telephone or Telegraph, trunk route	
Horizontal Control Point	△
Boundary Marker	□
Bench Mark	↑ BM 3025'
Spot Elevation, (in feet)	. 4582
Mine or Pit	✕

Streams	
Highways	--- ---
Roads	--- ---
Trails	
Houses	↑
Railroad	++++++
Falls	==
Rapids	==
Rip-Rap	^
Bridges	==

Dam	
Log Jams	
Log	
Power Line	--- ---
Coho	CO
Chum	CM
Pink	P
Chinook	CK
Sockeye	S

House, Building	•
School	⌘
Church	+
" with conspicuous Tower or Spire	⌘
Post Office	P
Tower, Radio Mast, Lookout, etc.	o
Cemetery	☠
Quarry	
Sand or Gravel Pit	
Cliff	
Cutting	
Embankment	
Dry River Bed	

Lighthouse	★
Wharf or Pier	
Foreshore Flats	
Swamp or Marsh	
Lake or Pond, intermittent	
Glacier or Snowfield	
Stream, intermittent, indefinite	
Irrigation Canals, Ditches	
Inundated Land, seasonal	
Contours, elevation	500 400
" depression	500 400
" approximate	500 400
Forest, unclassified	

Campsite	⊙
Provincial Park less than 40 Acres	⬆
Post Office	P
Settlement	•
School	⌘
Hospital	H
Mine	✕
Historic Monument	⊕
Forest Service Lookout	⬆
Surveyed land in flooded areas	
Micro Wave Relay Station	

Town	Ville	□	Stream	Cours d'eau
Village or Settlement	Village ou hameau	○	intermittent or dry	intermittent ou à sec
Post Office	Bureau de poste	P	Intermittent lake	Lac intermittent
Church	Église	⌘	Rapids; falls	Rapides; chute
School	École	⌘	Marsh or Swamp	Marais ou marécage
Boundary monument	Borne frontière	⊙	Lighthouse	Phare
Airport	Aéroport	⬆	Horizontal control point	Point géodésique
Seaplane base	Base d'hydravions	⬆	Landing ground	Piste d'atterrissage
			Seaplane anchorage	Amarrage d'hydravions

STREAM DATA
STATISTICAL AREA 19



NAME OF STREAM _____ (Ayum Creek)
 CONSERVATION DISTRICT 3 STATISTICAL AREA 19
 LOCATION OF MOUTH Flows SE. and SW. into Cooper Cove, N. side of Sooke Basin,
Sooke Dist. POSITION 48 123 SW.
 LENGTH 0.5 MI. WIDTH _____ FT. DRAINAGE _____ SQ. MI.
 COMPOSITION: BEDROCK _____ BOULDER _____ COARSE _____ FINE _____
 SILT & SAND _____ UNCLASSIFIED _____

GRADIENT:

FALL IN FT/000

☐ 0.0 - 2.5
☐ 2.5 - 5.0
☐ 5.0 - 7.5
☐ 7.5 - 10.0
☐ > 10.0

WETTED AREA _____ SQ. YD. SPAWNING AREA _____ SQ. YD.

DISCHARGE _____ CFS MAX _____ MIN _____

TEMPERATURE _____

BARRIERS OR POINTS OF DIFFICULT ASCENT _____

- Impassable falls at 0.75 mi.

SPAWNING DISTRIBUTION:

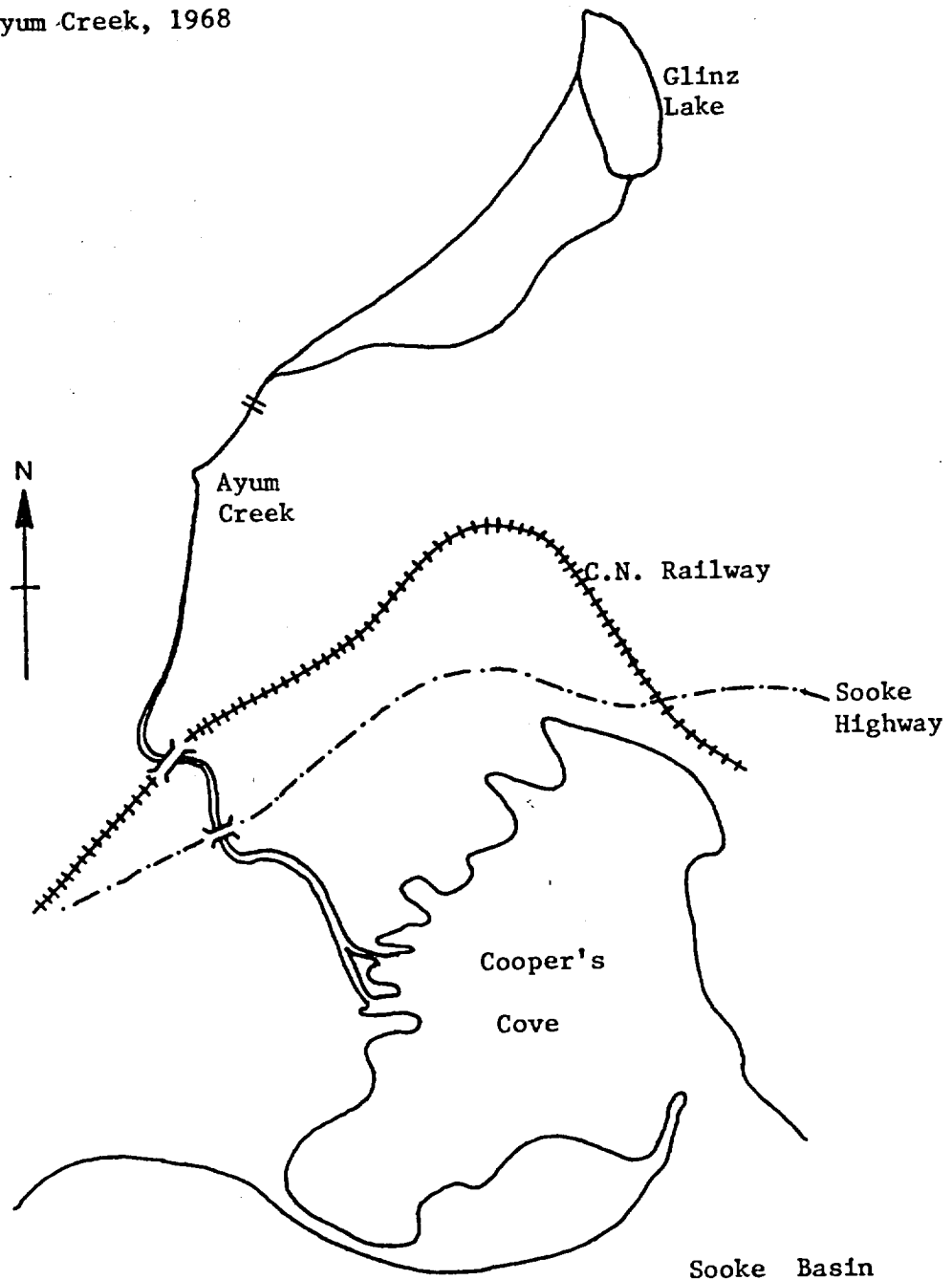
SPECIES	SECTION OF STREAM USED
SOCKEYE	
CHINOOK	
COHO	concentrated below railway bridge (0.5 mi.)
CHUM	concentrated below railway bridge (0.5 mi.)
PINK (ODD YR)	
PINK (EVEN YR)	
STEELHEAD	

POTENTIAL OF INACCESSIBLE PORTION OF STREAM _____

GENERAL REMARKS:

- Summer flow is largely dependant on a controlled make-shift dam at Glintz Lake. The dam was constructed by the Y.M.C.A. camp. (1962-1970)
- Molestation of fish is a problem. (1968)
- A sawmill and houses in Cooper's Cove draw water from this creek. (1970)
- Early chums were delayed one month in Sooke Basin due to low water. (1974)
- Frequent reports in the past have indicated over spawning in the lower reaches. (1974)
- An Indian food fishery took an estimated 15% of the chum run. (1975)

Sketch of Ayum Creek, 1968

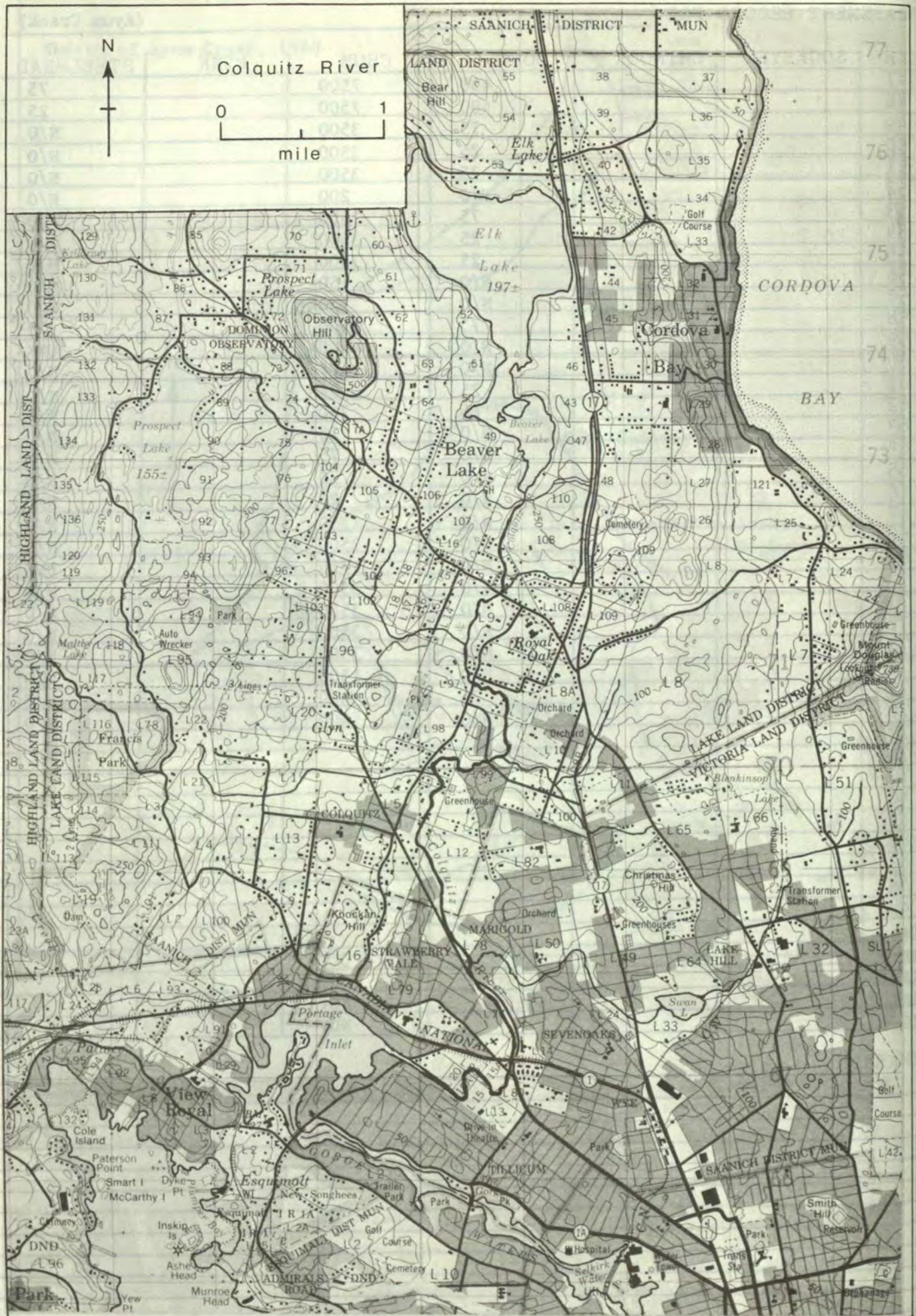


ESCAPEMENT RECORD FOR

(Ayum Creek)

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947			400	7500		75
148			400	7500		25
149			75	3500		N/O
150			25	3500		N/O
151			75	3500		N/O
152			N/O	200		N/O
153			75	3500		N/O
154			25	3500		N/O
155			25	3500		N/O
156			N/O	150		N/O
157			N/O	750		N/O
158			75	3500		25
159			N/O	200		UNK
160			N/O	25		N/O
161			25	25		N/O
162			25	750		N/O
163			N/O	25		N/O
164			N/O	75		N/O
165			N/O	750		N/O
166			N/O	25		N/O
167			N/O	750		N/O
168			N/O	3500		N/O
169			N/O	750		UNK
170			N/O	3500		N/O
171			N/O	3500		N/O
172			N/O	3500		N/O
173			N/O	3500		N/O
174			N/O	1500		N/O
175			25	400		N/O
176			N/O	400		N/O
177						
178						
179						
180						
181						
182						
183						
184						
185						
Time						
Arr.				L OCT		
Start				L OCT		
Peak				E NOV		
End				L NOV		

REMARKS



NAME OF STREAM COLQUITZ RIVER

CONSERVATION DISTRICT 3 STATISTICAL AREA 19

LOCATION OF MOUTH Flows S. and W. into Portage Inlet, Victoria Dist.

POSITION 48 123 SE.

LENGTH 4 MI. WIDTH FT. DRAINAGE SQ. MI.

COMPOSITION: BEDROCK BOULDER COARSE FINE

SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

10.0 - 2.5	
12.5 - 5.0	
15.0 - 7.5	
17.5 - 10.0	
> 10.0	

WETTED AREA SQ. YD. SPAWNING AREA SQ. YD.DISCHARGE CFS MAX MIN TEMPERATURE BARRIERS OR POINTS OF DIFFICULT ASCENT - Impassable falls at 4 mi.

SPAWNING DISTRIBUTION:

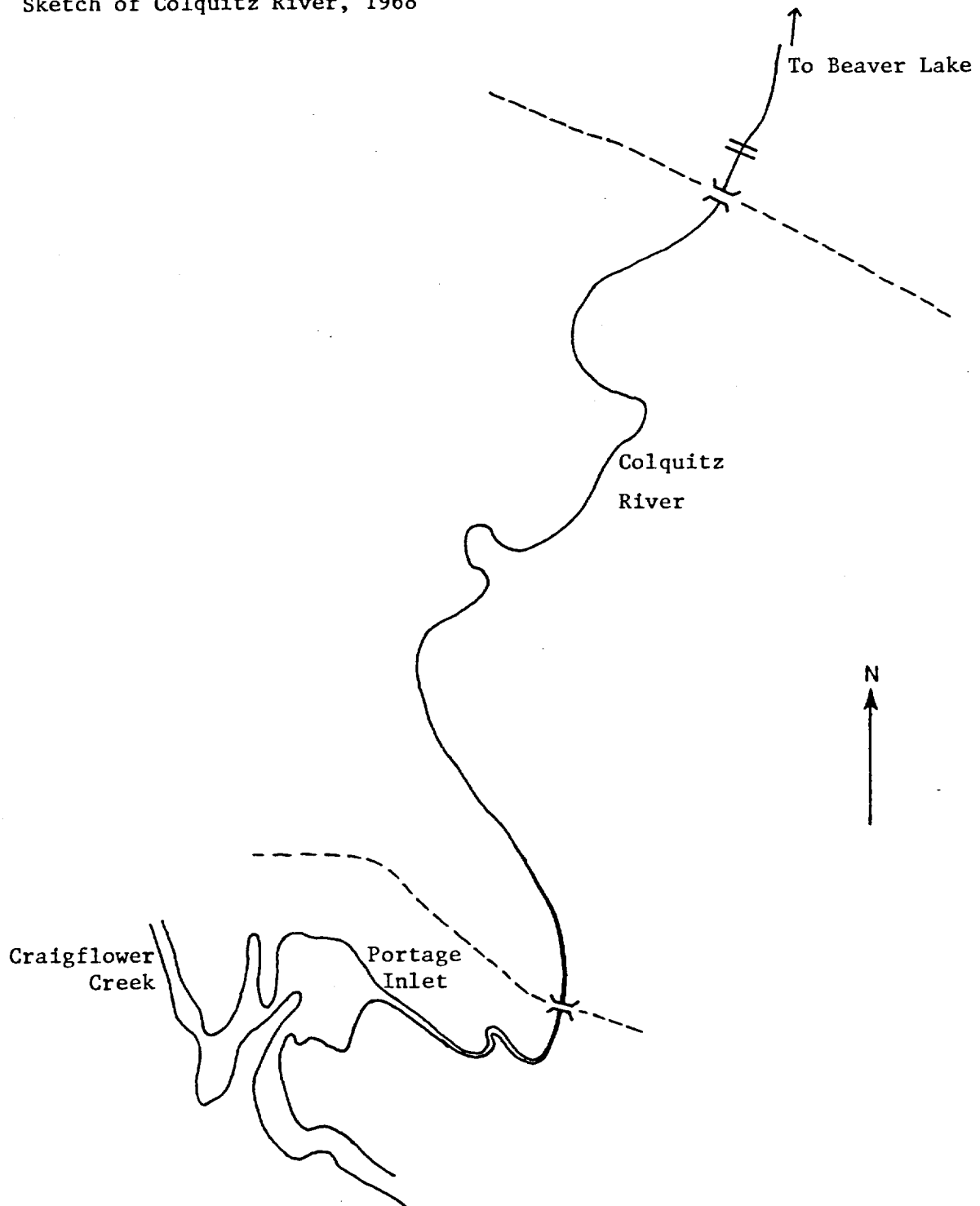
SPECIES	SECTION OF STREAM USED
SOCKEYE	
CHINOOK	
COHO	to falls
CHUM	
PINK (ODD YR)	
PINK (EVEN YR)	
STEELHEAD	

POTENTIAL OF INACCESSIBLE PORTION OF STREAM

GENERAL REMARKS:

- This stream is utilized, in residential areas, for sewage disposal. (1952)
- A concrete dam was installed at the outlet of Beaver Lake by the city of Victoria. The dam was constructed in conjunction with the Parks Board to enhance recreational potential around the lake and to provide water for North Saanich Municipality. (1962)
- Silting has occurred due to the farming community. (1965-66)
- The many water licences on this stream result in very low water levels during the summer and early fall. (1968-69)
- Heavy silting occurred from 1964-1971.
- Three weirs were constructed by the Saanich Municipality to enhance recreational potential along the Colquitz River. Both salmon and trout were observed passing through this area without difficulty. (1973)

Sketch of Colquitz River, 1968



ESCAPEMENT RECORD FOR COLQUITZ RIVER

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947			400	200		N/O
148			200	N/O		N/O
149			75	N/O		N/O
150			200	N/O		N/O
151			75	N/O		N/O
152			75	N/O		25
153						
154						
155						
156			NO RECORDS FOR 1953 - 1959			
157						
158						
159						
60			25	N/O		N/O
61			75	N/O		N/O
62			75	N/O		N/O
63				NO RECORDS		
64			N/O	N/O		N/O
65			75	N/O		N/O
66			75	N/O		N/O
67			N/O	N/O		N/O
68			75	N/O		N/O
69			200	N/O		N/O
70			N/O	N/O		N/O
71			N/O	N/O		N/O
72			75	N/O		N/O
73			25	N/O		N/O
74			15	N/O		N/O
75			75	N/O		N/O
76			25	N/O		N/O
77						
78						
79						
80						
81						
82						
83						
84						
85						
Time						
Arr.			M	NOV		
Start			E	DEC		
Peak			M	DEC		
End			L	DEC		

REMARKS



NAME OF STREAM CRAIGFLOWER CREEKCONSERVATION DISTRICT 3 STATISTICAL AREA 19LOCATION OF MOUTH Flows SE. into Portage Inlet, Esquimalt Dist.POSITION 48 123 SE.LENGTH MI. WIDTH FT. DRAINAGE SQ. MI.COMPOSITION: BEDROCK BOULDER COARSE FINE SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

10.0 - 2.512.5 - 5.015.0 - 7.517.5 - 10.0> 10.0WETTED AREA SQ. YD.SPAWNING AREA SQ. YD.DISCHARGE CFS MAX MIN TEMPERATURE BARRIERS OR POINTS OF DIFFICULT ASCENT - Falls at 2.5 mi. are passable at high water.

SPAWNING DISTRIBUTION:

SPECIES

SECTION OF STREAM USED

SCKEYE

CHINOOK

COHO

CHUM

PINK (ODD YR)

PINK (EVEN YR)

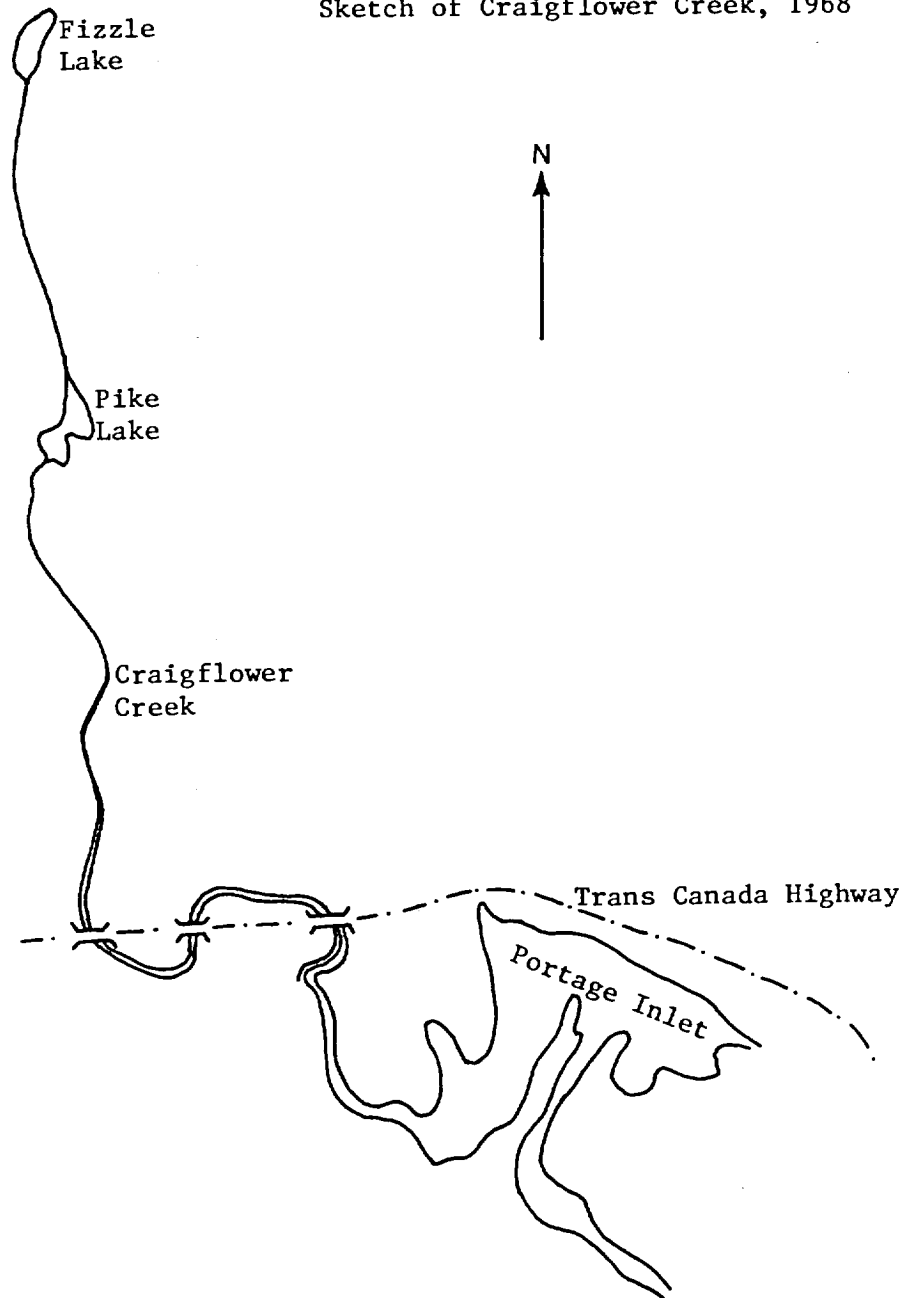
STEELHEAD

POTENTIAL OF INACCESSIBLE PORTION OF STREAM

GENERAL REMARKS:

- Small log jams were removed from the lower portion of the creek. (1974)

Sketch of Craigflower Creek, 1968



ESCAPEMENT RECORD FOR CRAIGFLOWER CREEK

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947			750	400		25
48			400	N/O		N/O
49			200	N/O		N/O
50			200	N/O		N/O
51			400	N/O		N/O
52			750	N/O		25
53			750	N/O		25
54			75	N/O		25
55			25	N/O		UNK
56			750	N/O		25
57			75	N/O		N/O
58			200	N/O		25
59			75	N/O		UNK
60			75	N/O		UNK
61			200	N/O		UNK
62			75	N/O		N/O
63			200	N/O		N/O
64			200	N/O		N/O
65			200	N/O		N/O
66			400	N/O		N/O
67			N/O	N/O		N/O
68			75	N/O		N/O
69			200	N/O		N/O
70			750	N/O		N/O
71			25	N/O		N/O
72			25	N/O		N/O
73			47	N/O		N/O
74			75	N/O		N/O
75			25	N/O		N/O
76			25	N/O		N/O
77						
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83						
84						
85						
Time						
Arr.			E DEC			
Start			E DEC			
Peak			M DEC			
End			L DEC			

REMARKS



NAME OF STREAM GOLDSTREAM RIVERCONSERVATION DISTRICT 3 STATISTICAL AREA 19LOCATION OF MOUTH Flows SE. and N. into head of Finlayson Arm, Goldstream Dist.POSITION 48 123 SW.LENGTH 2.5 MI. WIDTH FT. DRAINAGE SQ. MI.COMPOSITION: BEDROCK BOULDER COARSE FINE SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

0.0 - 2.52.5 - 5.05.0 - 7.57.5 - 10.0> 10.0WETTED AREA SQ. YD.SPAWNING AREA SQ. YD.DISCHARGE CFS MAX MIN TEMPERATURE BARRIERS OR POINTS OF DIFFICULT ASCENT - Passable falls at 1.5 mi.- Impassable falls at 2.5 mi. (1975)

SPAWNING DISTRIBUTION:

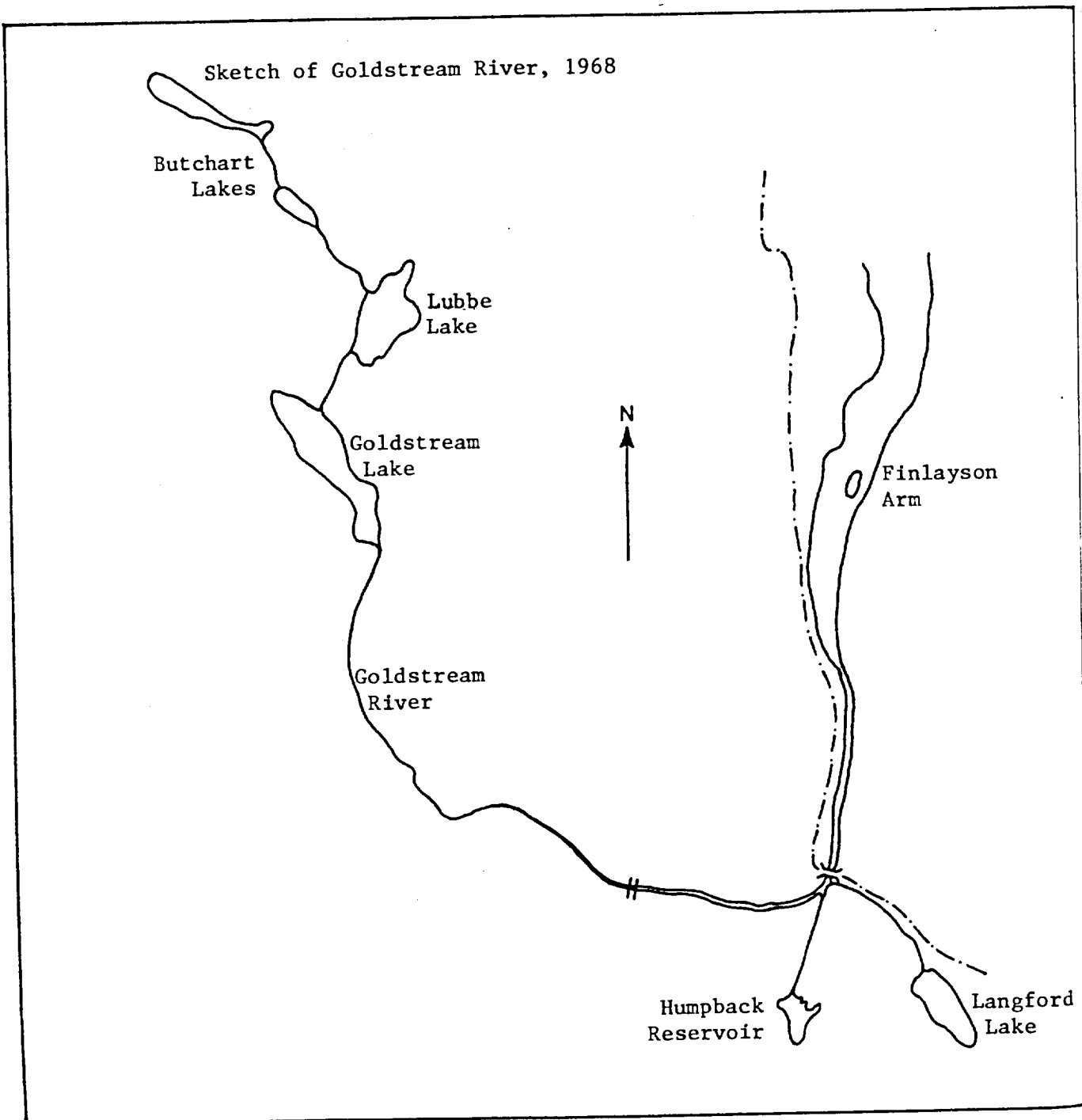
SPECIES SECTION OF STREAM USED SCKEYE CHINOOK COHO throughout to fallsCHUM PINK (ODD YR) PINK (EVEN YR) STEELHEAD POTENTIAL OF INACCESSIBLE PORTION OF STREAM

GENERAL REMARKS:

- Flood conditions in Jan. scoured the spawning beds resulting in a heavy loss to salmon eggs. Some coho fry were also lost. (1960)
- Summer flow was aided with co-operation from the Victoria Water Board. (1962-66)
- Dams were constructed to save spawning beds from drying out and freezing. (1969)
- The Goldstream River watershed, in conjunction with the Sooke River watershed, supply water to the greater Victoria area. The flow is controlled by a number of dams on tributary lakes and on Goldstream Lake. (1970)
- Over 50,000 people visited this stream to observe spawning salmon. Salmon were unable to ascend the river until water was released by the Victoria Water Board. (1970)

GENERAL REMARKS: (Cont'd)

- A fair water flow was provided by the Victoria Water Board until early October when not enough water was provided to accommodate salmon. (1972)
- An incubation box was installed by the Amalgamated Conservation Society in the vicinity of Japan Gulch Dam. A total of 50,250 coho eggs were planted. (1974)
- The Provincial Parks Branch conducted some stream improvement work on the lower reaches. (1974)



ESCAPEMENT RECORD FOR GOLDSTREAM RIVER

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947		2	1500	15000		75
48		5	1500	35000		75
49		N/O	750	35000		25
50		N/O	1500	35000		25
51		N/O	1500	15000		UNK
52		N/O	1500	35000		25
53		N/O	200	7500		25
54		N/O	400	15000		25
55		N/O	75	3500		UNK
56		N/O	200	3500		25
57		N/O	75	7500		25
58		N/O	200	7500		25
59		N/O	400	3500		UNK
60		N/O	500	2200		UNK
61		N/O	400	7500		UNK
62		25	75	7500		UNK
63		2	400	3500		UNK
64		UNK	2100	5500		UNK
65		25	100	12000		N/O
66		25	615	9800		UNK
67		40	420	7000		25
68		N/O	542	7000		N/O
69		23	218	8000		UNK
70		25	1500	7500		N/O
71		25	1500	3500		UNK
72		25	200	3500		N/O
73		25	200	7500		N/O
74		25	578	7500		N/O
75		75	200	1500		N/O
76		25	200	3500		N/O
77						
78						
79						
80						
81						
82						
83						
84						
85						
Time						
Arr.		E OCT	L OCT	M OCT		
Start		E OCT	L OCT	L OCT		
Peak		L OCT	L OCT	M NOV		
End		E DEC	L DEC	L DEC		

REMARKS



NAME OF STREAM SANDHILL CREEK (Shady Creek)
 CONSERVATION DISTRICT 3 STATISTICAL AREA 19
 LOCATION OF MOUTH Flows N. and NE. into Saanichton Bay, South Saanich Dist.

LENGTH MI. WIDTH FT. POSITION 48 123 NE.
 DRAINAGE SQ. MI.
 COMPOSITION: BEDROCK BOULDER COARSE FINE
 SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

10.0 - 2.5
12.5 - 5.0
15.0 - 7.5
17.5 - 10.0
> 10.0

WETTED AREA SQ. YD. SPAWNING AREA SQ. YD.

DISCHARGE CFS MAX MIN

TEMPERATURE BARRIERS OR POINTS OF DIFFICULT ASCENT

- Passable windfalls throughout. (1975)

- Impassable dam at 2 mi.

SPAWNING DISTRIBUTION:

SPECIES

SECTION OF STREAM USED

SCKEYE

CHINOOK

COHO

throughout

CHUM

PINK (ODD YR)

PINK (EVEN YR)

STEELHEAD

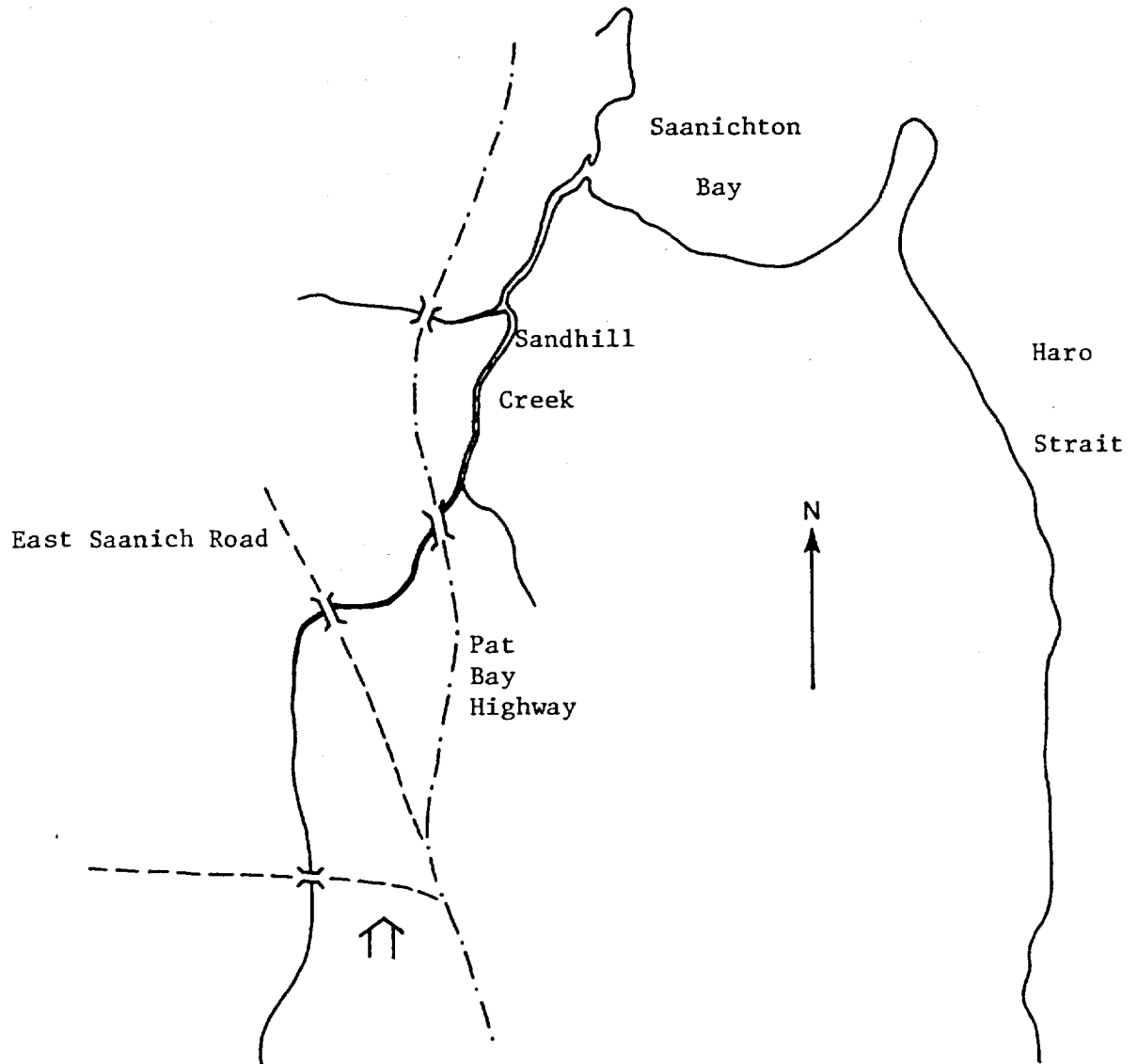
POTENTIAL OF INACCESSIBLE PORTION OF STREAM

GENERAL REMARKS:

- Old logging debris was removed. (1970)

- A stream clearing program was carried out by the National Parks Board. (1975)

Sketch of Sandhill Creek, 1968



ESCAPEMENT RECORD FOR SANDHILL CREEK

(Shady Creek)

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947						
48						
49						
50						
51						
52						
53						
54						
55						
56						
57						
58						
59						
60						
61						
62						
63			NO RECORDS PRIOR TO 1965			
64						
65			25			
66			75			
67			N/O			
68			25			
69			25			
70			N/O			
71			N/O			
72			N/O			
73			N/O			
74			N/O			
75			20			
76			75			
77			N/O			
78						
79						
80						
81						
82						
83						
84						
85						
Time						
Arr.						
Start						
Peak						
End						

REMARKS



NAME OF STREAM SHAWNIGAN CREEK (Mill Bay Creek)
 CONSERVATION DISTRICT 3 STATISTICAL AREA 19
 LOCATION OF MOUTH Flows N. through Shawnigan L., then SE. & E. into Mill Bay.
Shawnigan Dist. POSITION 48 123 NW.
 LENGTH .25 MI. WIDTH FT. DRAINAGE SQ. MI.
 COMPOSITION: BEDROCK BOULDER COARSE FINE
 SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

10.0 - 2.5	
12.5 - 5.0	
15.0 - 7.5	
17.5 - 10.0	
1 > 10.0	

WETTED AREA SQ. YD. SPAWNING AREA SQ. YD.

DISCHARGE CFS MAX MIN

TEMPERATURE

BARRIERS OR POINTS OF DIFFICULT ASCENT

- Impassable falls at .25 mi.

SPAWNING DISTRIBUTION:

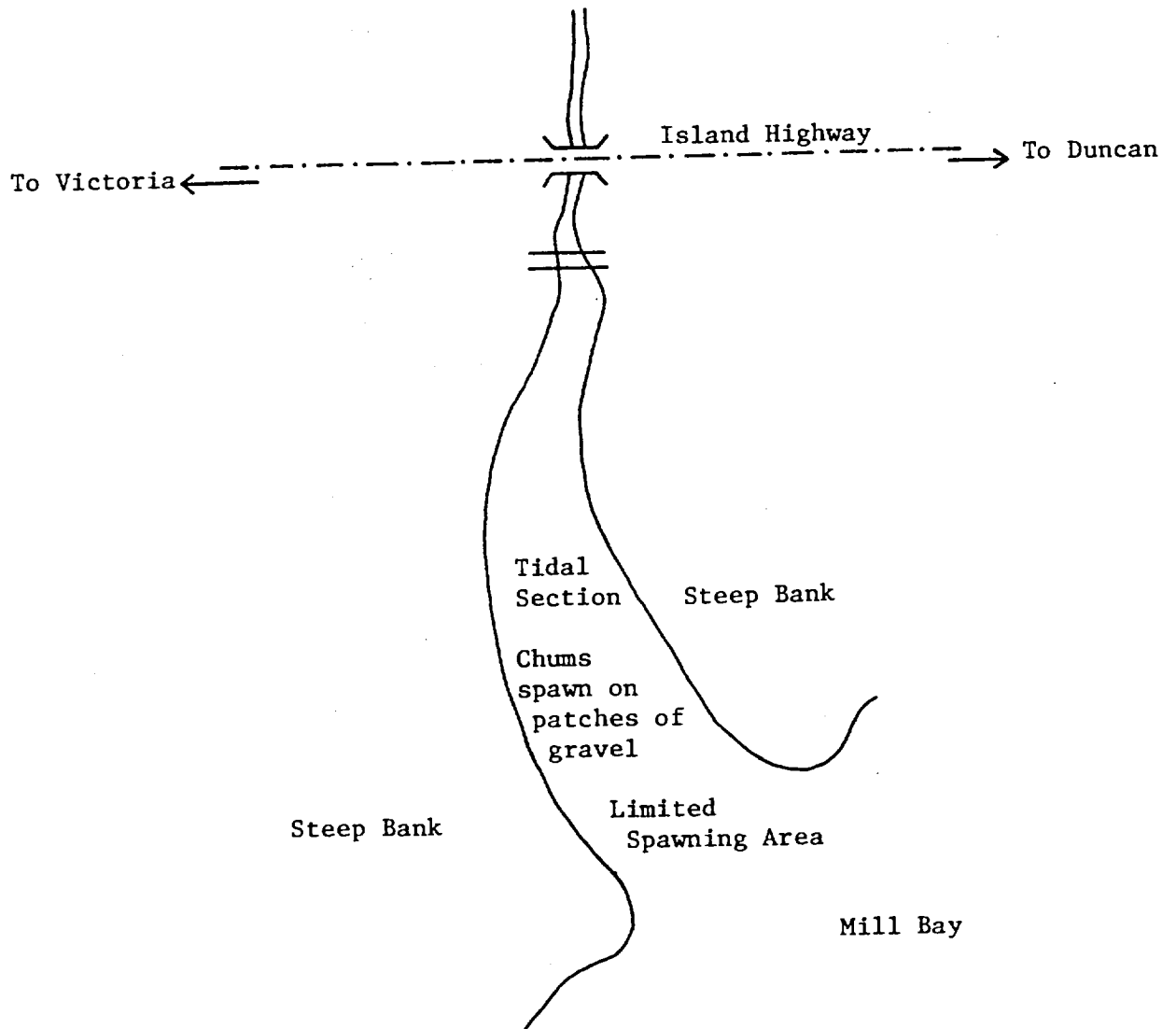
SPECIES	SECTION OF STREAM USED
SOCKEYE	
CHINOOK	
COHO	tidal section
CHUM	tidal section
PINK (ODD YR)	
PINK (EVEN YR)	
STEELHEAD	

POTENTIAL OF INACCESSIBLE PORTION OF STREAM

GENERAL REMARKS:

- There is a very limited spawning area in this stream. (1970)
 - In 1969, fry transplanting above the falls was introduced and was discontinued in 1970. The cycle was completed in 1972.

Sketch of Shawnigan Creek, 1968
(Lower Portion)



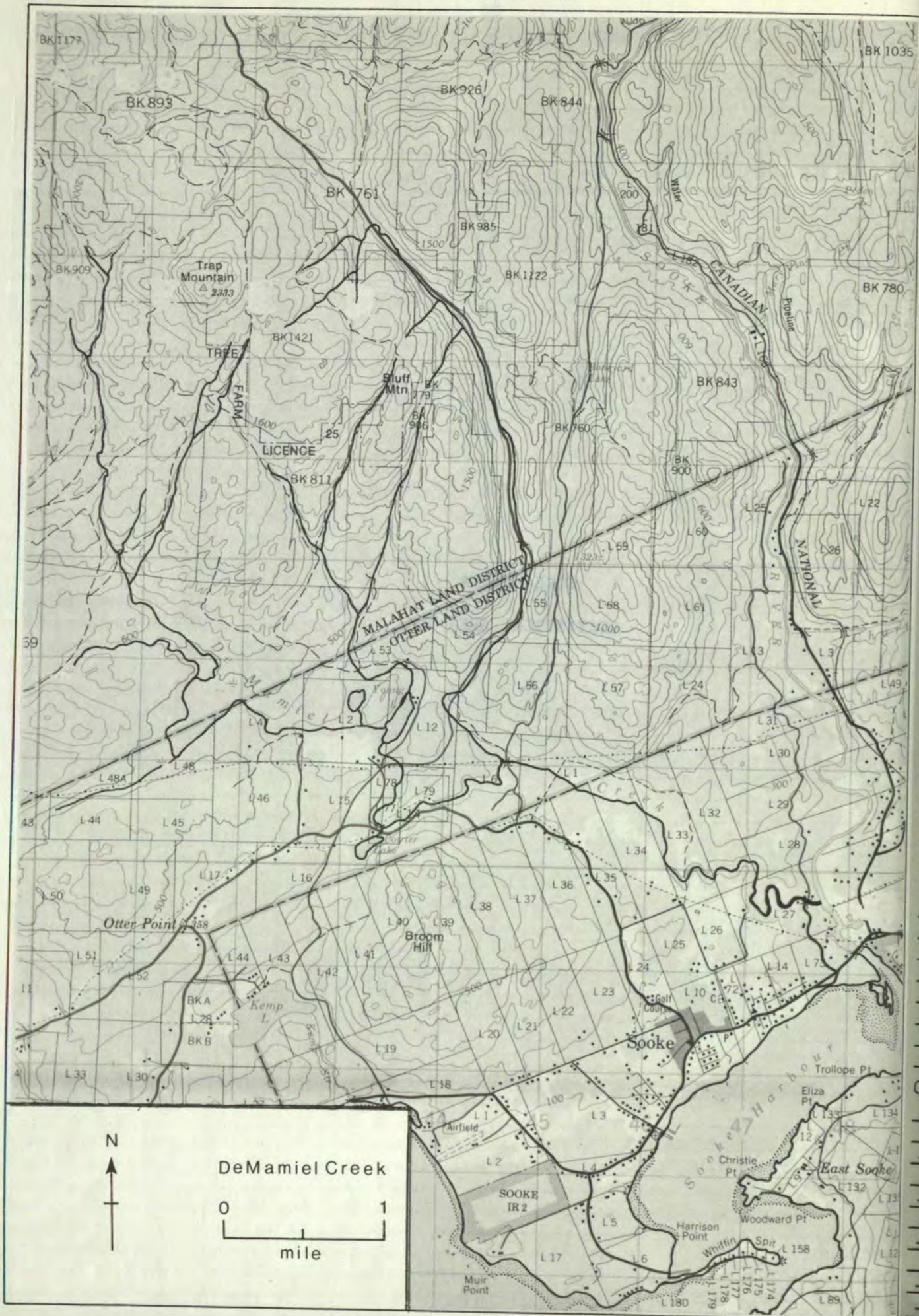
ESCAPEMENT RECORD FOR SHAWNIGAN CREEK

(Mill Bay Creek)

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947						
48						
49						
50						
51			NO RECORDS PRIOR TO 1954			
52						
53						
54			N/O	7500		
55			N/O	3500		
56			N/O	3500		
57			N/O	1500		
58			N/O	1500		
59			N/O	400		
60			N/O	75		
61			N/O	25		
62			N/O	200		
63			N/O	25		
64			N/O	75		
65			N/O	75		
66			N/O	200		
67			N/O	25		
68			N/O	200		
69			N/O	200		
70			N/O	75		
71			200	N/O		
72			25	N/O		
73			N/O	N/O		
74						
75			NO RECORDS FOR 1974 - 1976			
76						
77						
78						
79						
80						
81						
82						
83						
84						
85						
Time						
Arr.				E	NOV	
Start				M	NOV	
Peak				L	NOV	
End				M	DEC	

REMARKS

STREAM DATA
STATISTICAL AREA 20



NAME OF STREAM DeMAMIEL CREEKCONSERVATION DISTRICT 3 STATISTICAL AREA 20LOCATION OF MOUTH Flows SE. into Sooke R. near mouth, Sooke Dist.POSITION 48 123 SW.LENGTH MI. WIDTH FT. DRAINAGE SQ. MI.COMPOSITION: BEDROCK BOULDER COARSE FINE SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

0.0 - 2.52.5 - 5.05.0 - 7.57.5 - 10.0> 10.0WETTED AREA SQ. YD.SPAWNING AREA SQ. YD.DISCHARGE CFS MAX MIN TEMPERATURE BARRIERS OR POINTS OF DIFFICULT ASCENT - Passable falls at 3.6 and 5.7 mi.

SPAWNING DISTRIBUTION:

SPECIES

SECTION OF STREAM USED

SCKEYE

CHINOOK

COHO

throughout

CHUM

lower reaches (some years to first falls)

PINK (ODD YR)

PINK (EVEN YR)

STEELHEAD

POTENTIAL OF INACCESSIBLE PORTION OF STREAM

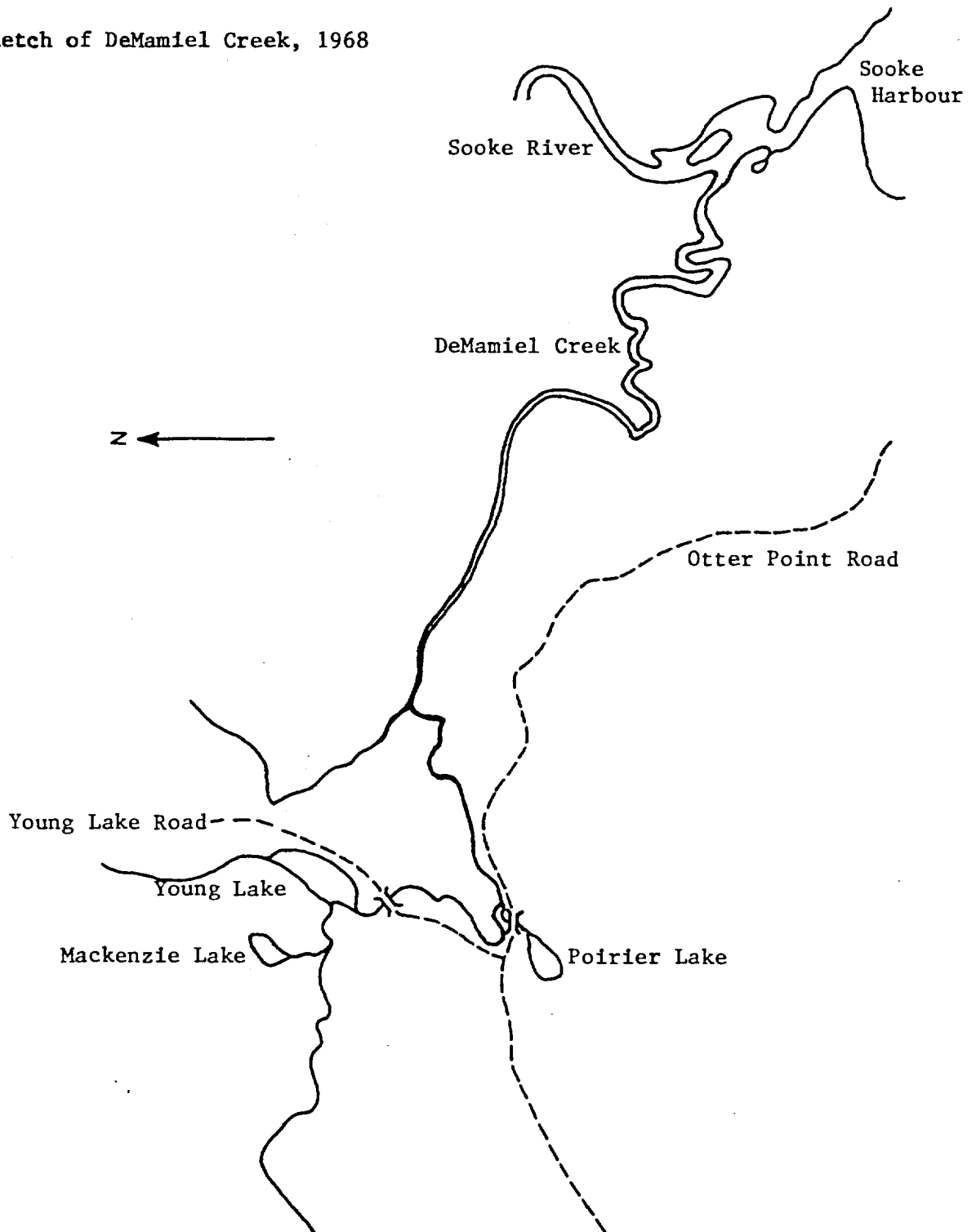
GENERAL REMARKS:

- It has been 15 years since this watershed was logged and a good stand of second growth timber presently exists. (1950)
- The drop in coho & chum spawning was partly caused by the loss of eggs which were washed out in 1955 (chums) and 1956 (coho) floods. (1959)
- Flood conditions scoured spawning beds which resulted in a great loss to eggs, especially chum. (1961)
- November floods caused chum mortality. Make-shift dams at McKenzie & Young's Lakes are believed to be the cause. (1962)
- This stream could double its output with a controlled flow. (1966)
- This stream requires more water in summer months for coho fry. (1968)

GENERAL REMARKS: (Cont'd)

- In the spring of 1970 the Fisheries Service constructed a storage dam at Young Lake in order to ensure a minimum flow of 0.5 cfs throughout the summer months. (1970)
- Due to impassable water levels much of DeMamiel Creek chum stock spawned in the lower Sooke River. (1974).

Sketch of DeMamiel Creek, 1968



ESCAPEMENT RECORD FOR DeMAMIEL CREEK

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947		N/O	3500	7500		200
148		N/O	3500	7500		200
149		N/O	1500	3500		75
150		N/O	1500	15000		25
151		N/O	1500	15000		75
152		N/O	750	750		25
153		N/O	400	15000		25
154		N/O	1500	7500		25
155		N/O	75	15000		25
156		N/O	400	7500		N/O
157		N/O	400	7500		25
158		N/O	3500	35000		25
159		N/O	750	7500		25
60		N/O	200	3500		N/O
61		N/O	200	3500		UNK
62		N/O	3500	3500		UNK
63		N/O	3500	3500		UNK
64		N/O	1500	15000		N/O
65		N/O	3500	7500		N/O
66		N/O	7500	7500		UNK
67		N/O	750	7500		N/O
68		N/O	3500	35000		N/O
69		N/O	750	15000		UNK
70		N/O	3500	15000		200
71		N/O	1500	7500		N/O
72		N/O	3500	15000		N/O
73		N/O	6400	35000		UNK
74		2	1500	3500		N/O
75		N/O	750	3500		N/O
76		N/O	400	1500		N/O
77						
78						
79						
80						
81						
82						
83						
84						
85						
Time						
Arr.			E NOV	L OCT		
Start			E NOV	L OCT		
Peak			L NOV	M NOV		
End			E DEC	E DEC		

REMARKS



NAME OF STREAM GORDON RIVER
CONSERVATION DISTRICT 3 STATISTICAL AREA 20
LOCATION OF MOUTH Flows SW. and S. into Port San Juan, Renfrew Dist.

POSITION 48 124 NE.
LENGTH MI. WIDTH FT. DRAINAGE SQ. MI.
COMPOSITION: BEDROCK BOULDER COARSE FINE
SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

0.0 - 2.5	
2.5 - 5.0	
5.0 - 7.5	
7.5 - 10.0	
> 10.0	

WETTED AREA SQ. YD. SPAWNING AREA SQ. YD.

DISCHARGE CFS MAX MIN

TEMPERATURE

BARRIERS OR POINTS OF DIFFICULT ASCENT

- Impassable falls at 8-10 mi.

SPAWNING DISTRIBUTION:

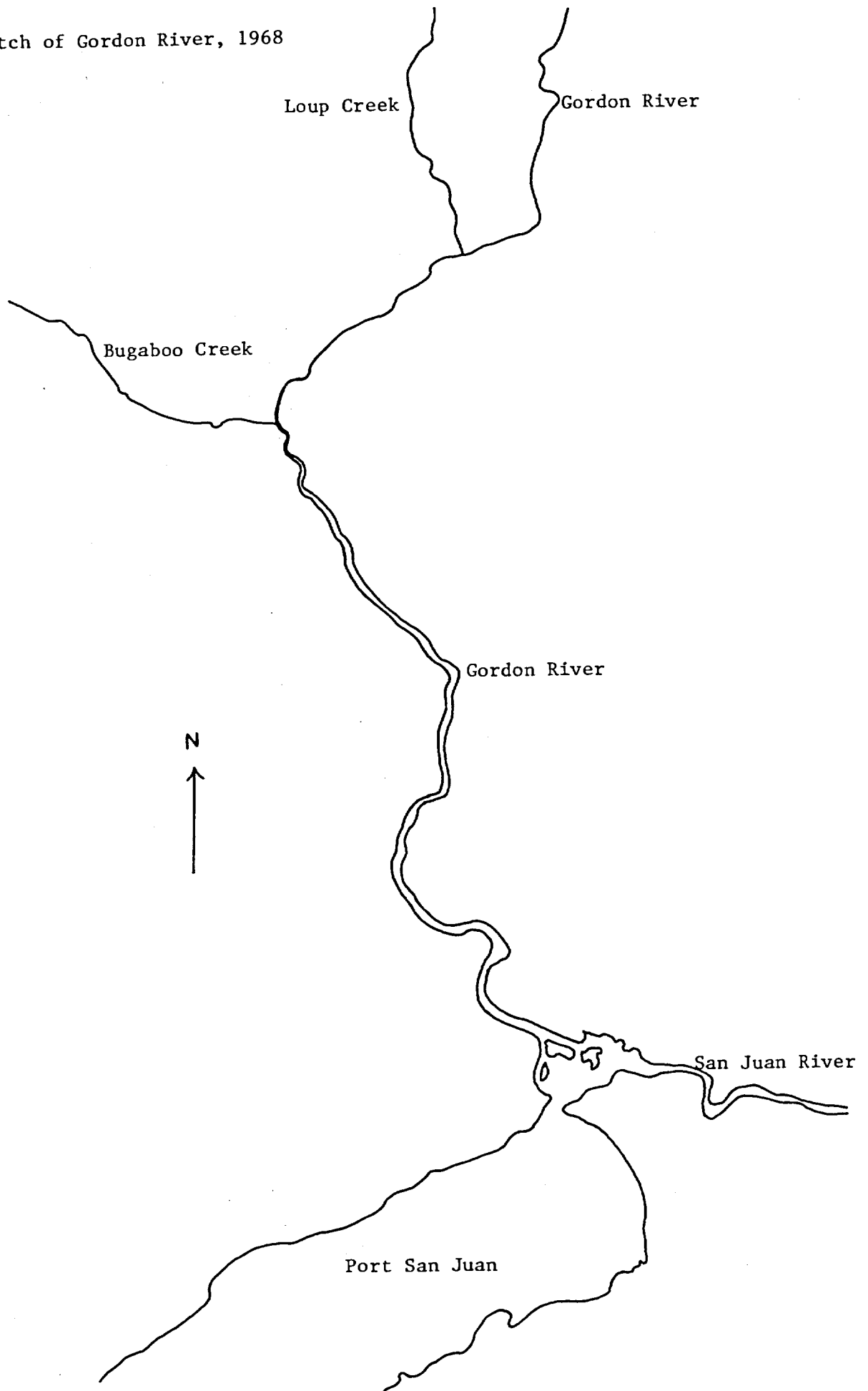
SPECIES	SECTION OF STREAM USED
SOCKEYE	
CHINOOK	
COHO	
CHUM	
PINK (ODD YR)	
PINK (EVEN YR)	
STEELHEAD	

POTENTIAL OF INACCESSIBLE PORTION OF STREAM

GENERAL REMARKS:

- This stream has very good gravel and the watershed is still mostly unlogged. (1955)
- The upper reaches of this stream are now being logged. (1970)

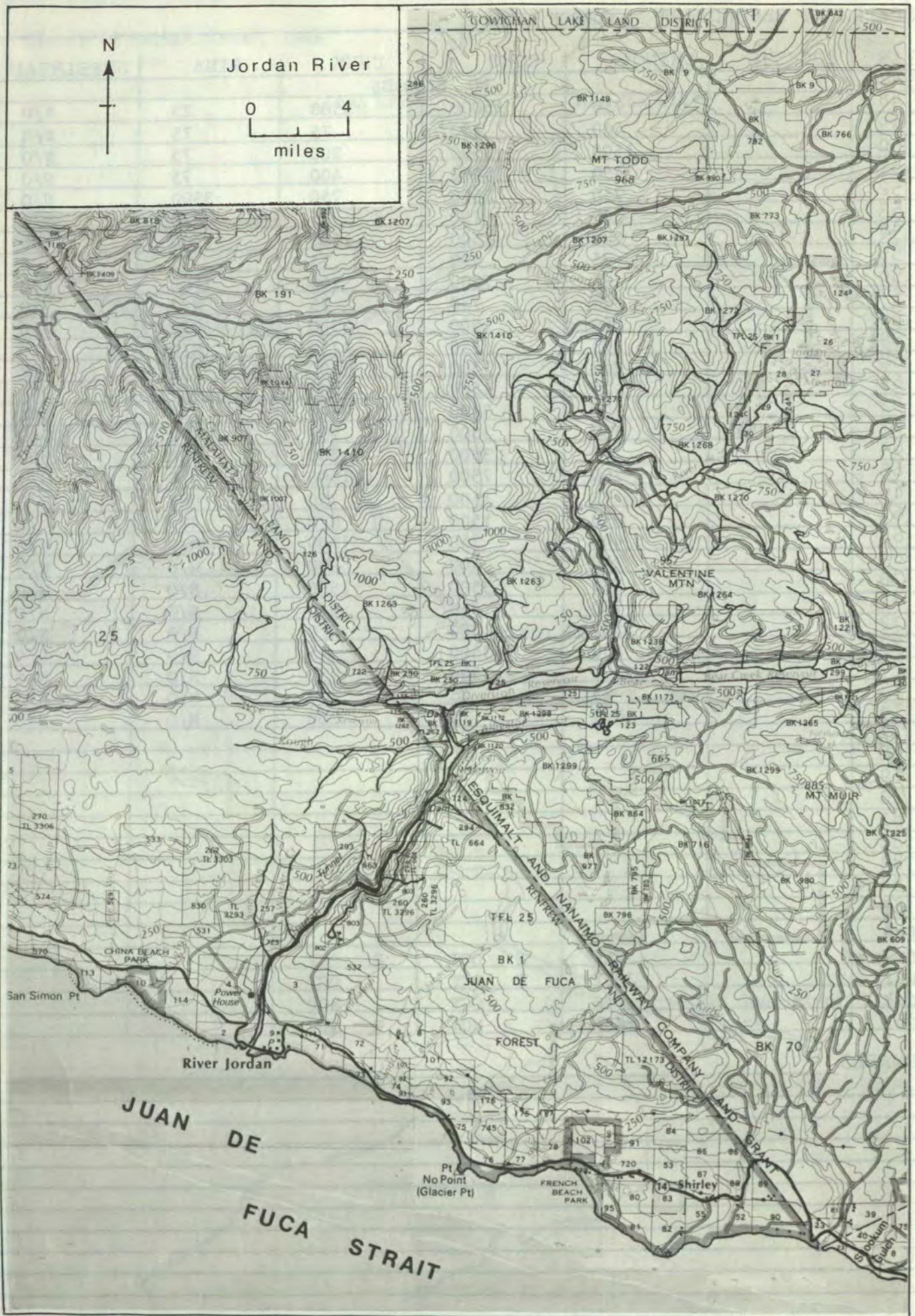
Sketch of Gordon River, 1968



ESCAPEMENT RECORD FOR GORDON RIVER

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947			NO RECORDS			
48		750	15000	3500	25	N/O
49		400	750	75	75	N/O
50		1500	3500	200	75	N/O
51		750	15000	400	75	N/O
52		750	3500	750	3500	N/O
53		200	750	75	N/O	N/O
54		200	1500	1500	1500	400
55		75	400	75	75	75
56		25	7500	200	1500	400
57		25	750	75	N/O	750
58		200	1500	1500	750	750
59		75	750	750	N/O	750
60		200	7500	1500	750	UNK
61		75	1500	1500	N/O	UNK
62	1	200	3500	3500	750	UNK
63	UNK	750	3500	750	N/O	3500
64		200	3500	200	UNK	1500
65		200	7500	1500	N/O	N/O
66		75	1500	750	UNK	N/O
67		750	1500	N/O	400	75
68		400	750	1500	N/O	N/O
69		200	400	200	N/O	N/O
70		750	3500	1500	N/O	N/O
71		750	750	N/O	N/O	N/O
72		3500	1500	N/O	N/O	N/O
73		400	3500	75	N/O	N/O
74		200	1500	750	N/O	N/O
75		25	25	200	N/O	N/O
76		N/O	75	200	N/O	N/O
77						
78						
79						
80						
81						
82						
83						
84						
85						
Time						
Arr.		JULY	E OCT	E OCT		
Start		JULY	E OCT	OCT		
Peak		AUG	OCT	NOV		
End		SEPT	NOV	E NOV		

REMARKS



NAME OF STREAM JORDAN RIVER

36

CONSERVATION DISTRICT 3STATISTICAL AREA 20LOCATION OF MOUTH Flows SW. into Juan de Fuca Str., SE. of San Simon Pt.,Renfrew Dist.POSITION 48 124 SE.LENGTH MI.WIDTH FT.DRAINAGE SQ. MI.COMPOSITION: BEDROCK BOULDER COARSE FINE SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

0.0 - 2.52.5 - 5.05.0 - 7.57.5 - 10.0> 10.0WETTED AREA SQ. YD.SPAWNING AREA SQ. YD.DISCHARGE CFS MAX MINTEMPERATURE BARRIERS OR POINTS OF DIFFICULT ASCENT - Impassable falls at 1.5 mi.

SPAWNING DISTRIBUTION:

SPECIES

SECTION OF STREAM USED

Sockeye

Chinook

Coho

Chum

Pink (odd yr)

Pink (even yr)

Steelhead

POTENTIAL OF INACCESSIBLE PORTION OF STREAM

GENERAL REMARKS:

- B.C. Hydro has used the Jordan River extensively for power development since about 1912. The project involves two storage dams, a power house near tidewater and a diversion flume and pipeline to the power house.

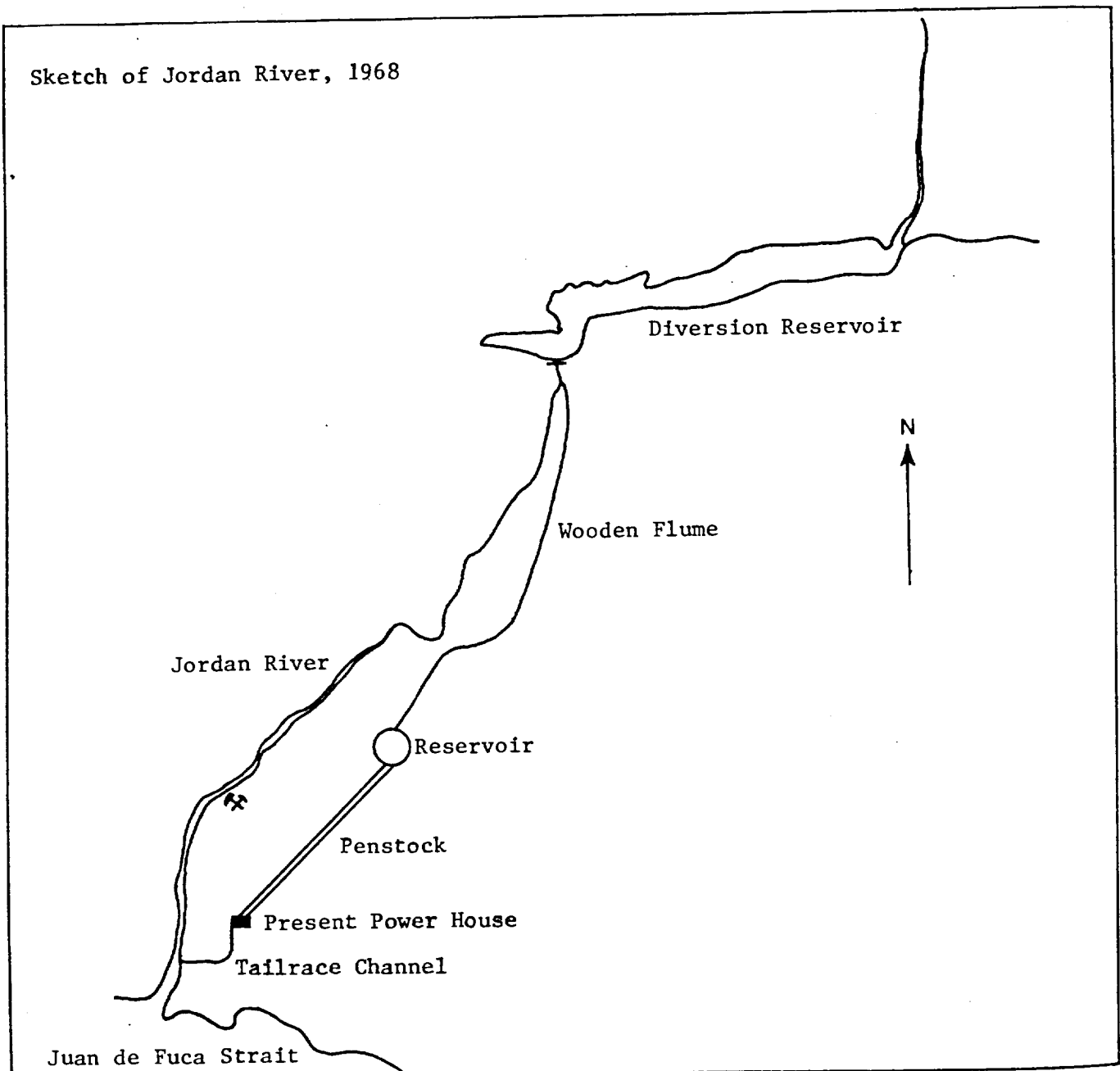
- In 1962, discharge flow from the power house through a tailrace channel was not consistent. This inconsistency of flow was detrimental since pink salmon use this channel for spawning. (1962)

- On Dec. 6 and again on Jan. 6, the river changed course twice due to collapsing mining tunnels. Hundreds of tons of silt and gravel poured into the lower river with silt and gravel deposited from a few inches to 9 ft. in depth. (1963)

GENERAL REMARKS: (Cont'd)

- B.C. Hydro agreed to provide 35 cfs which provides enough water for spawning purposes. (1966)
- A slide occurred in the canyon which took nearly a month to remove. (1966)
- There is no doubt hydro, logging, booming and mining have desimated the salmon runs. (1967)
- A third dam (Elliot Dam) and a new power house were completed in 1971. With the advent of these constructions this river is rendered useless for salmon production. (1971)

Sketch of Jordan River, 1968



ESCAPEMENT RECORD FOR JORDAN RIVER

38

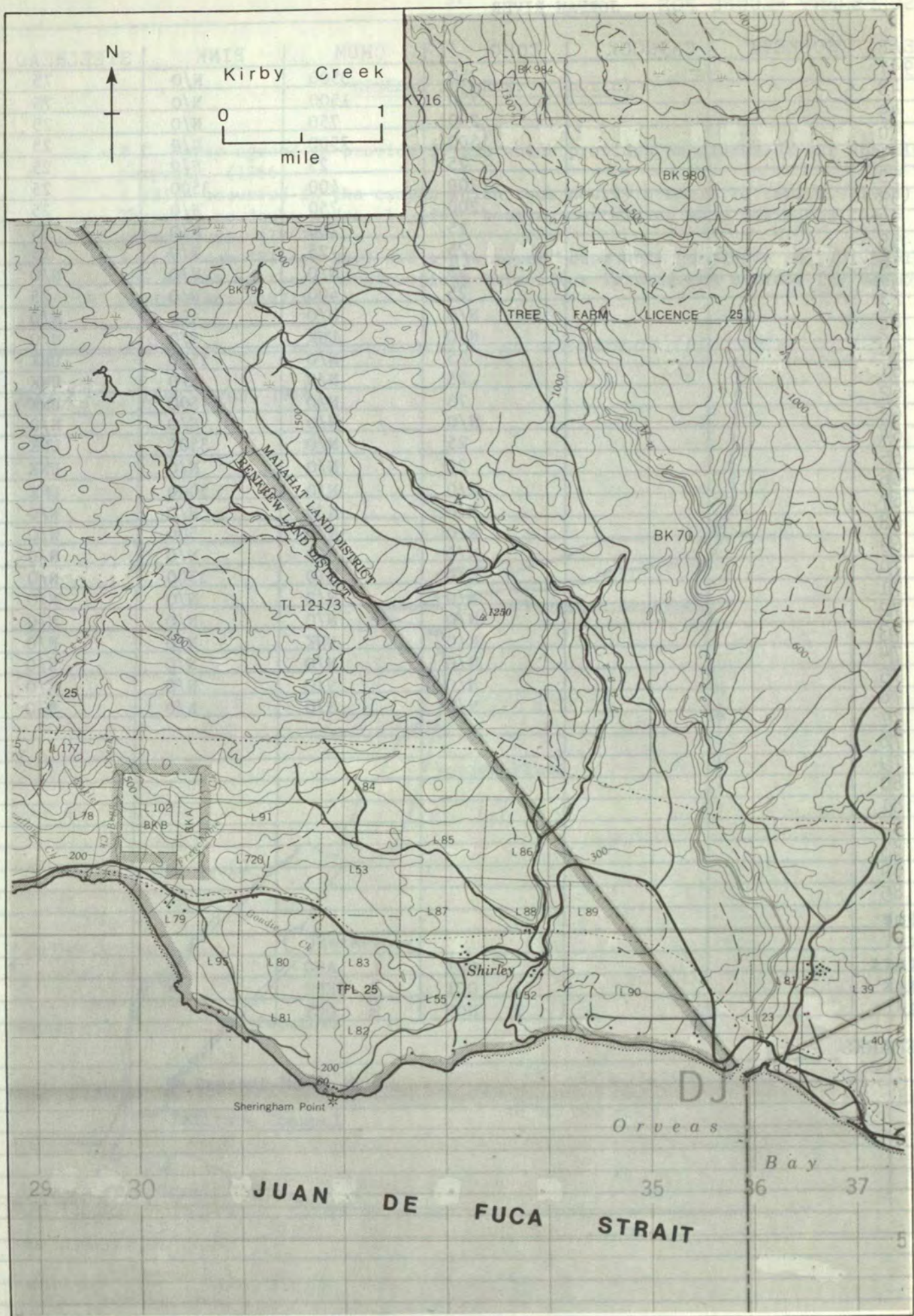
YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947						
48			750	3500	N/O	75
49			750	1500	N/O	75
50			200	750	N/O	25
51			1500	7500	N/O	25
52			25	25	N/O	25
53			200	400	3500	25
54			200	750	N/O	25
55			25	200	N/O	25
56			25	25	N/O	25
57			25	N/O	N/O	N/O
58			25	400	N/O	25
59			N/O	N/O	400	N/O
60			N/O	UNK	N/O	N/O
61			25	N/O	1500	UNK
62			25	N/O	N/O	UNK
63			75	N/O	7500	UNK
64			N/O	N/O	N/O	N/O
65			25	N/O	3500	UNK
66			25	N/O	N/O	UNK
67			N/O	N/O	5000	UNK
68			N/O	N/O	N/O	N/O
69			N/O	N/O	500	N/O
70			N/O	N/O	N/O	N/O
71			N/O	N/O	1500	N/O
72			N/O	N/O	N/O	N/O
73			N/O	N/O	N/O	N/O
74			N/O	UNK	N/O	N/O
75			N/O	N/O	N/O	N/O
76			N/O	N/O	N/O	N/O
77			N/O	N/O	N/O	N/O
78						
79						
80						
81						
82						
83						
84						
85						
Time						
Arr.						
Start						
Peak			OCT	L AUG		
End			L OCT	S SEPT		
			L NOV	M OCT		

REMARKS

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6)

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NAME OF STREAM KIRBY CREEK (Coal Creek)
 CONSERVATION DISTRICT 3 STATISTICAL AREA 20
 LOCATION OF MOUTH Flows S. into Orveas Bay, E. of Sheringham Pt., Renfrew Dist.
 LENGTH MI. WIDTH FT. POSITION 48 123 SW.
 COMPOSITION: BEDROCK BOULDER DRAINAGE SQ. MI.
 SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

0.0 - 2.5
2.5 - 5.0
5.0 - 7.5
7.5 - 10.0
> 10.0

WETTED AREA SQ. YD. SPAWNING AREA SQ. YD.
 DISCHARGE CFS MAX MIN
 TEMPERATURE

BARRIERS OR POINTS OF DIFFICULT ASCENT
- Impassable falls at 4 mi.

SPAWNING DISTRIBUTION:

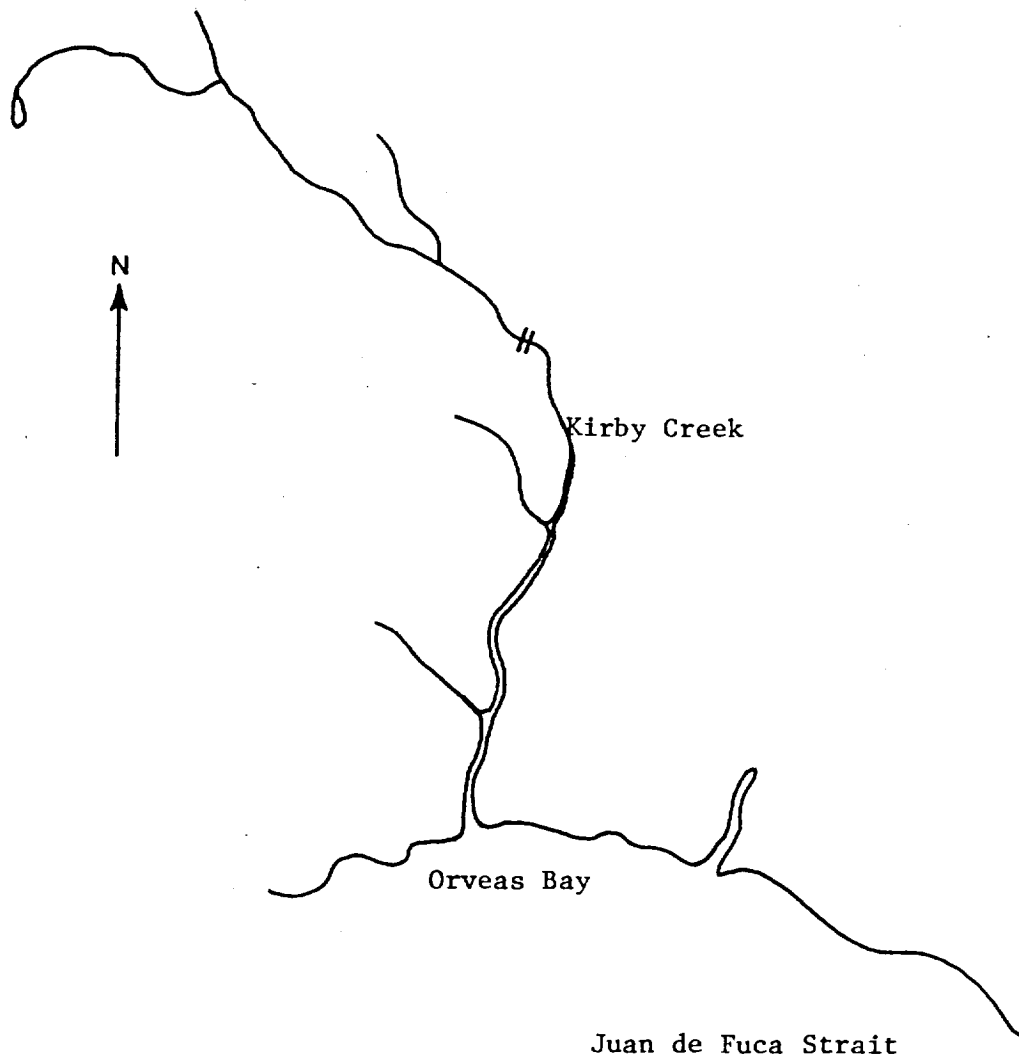
SPECIES	SECTION OF STREAM USED
SOCKEYE	
CHINOOK	
COHO	
CHUM	lower 0.5 mi.
PINK (ODD YR)	
PINK (EVEN YR)	
STEELHEAD	

POTENTIAL OF INACCESSIBLE PORTION OF STREAM

GENERAL REMARKS:

- This stream has stable flows and good gravel and should support larger runs. (1974)
- A channel was constructed at 1.5 mi. to bypass a log jam. (1975)

Sketch of Kirby Creek, 1968



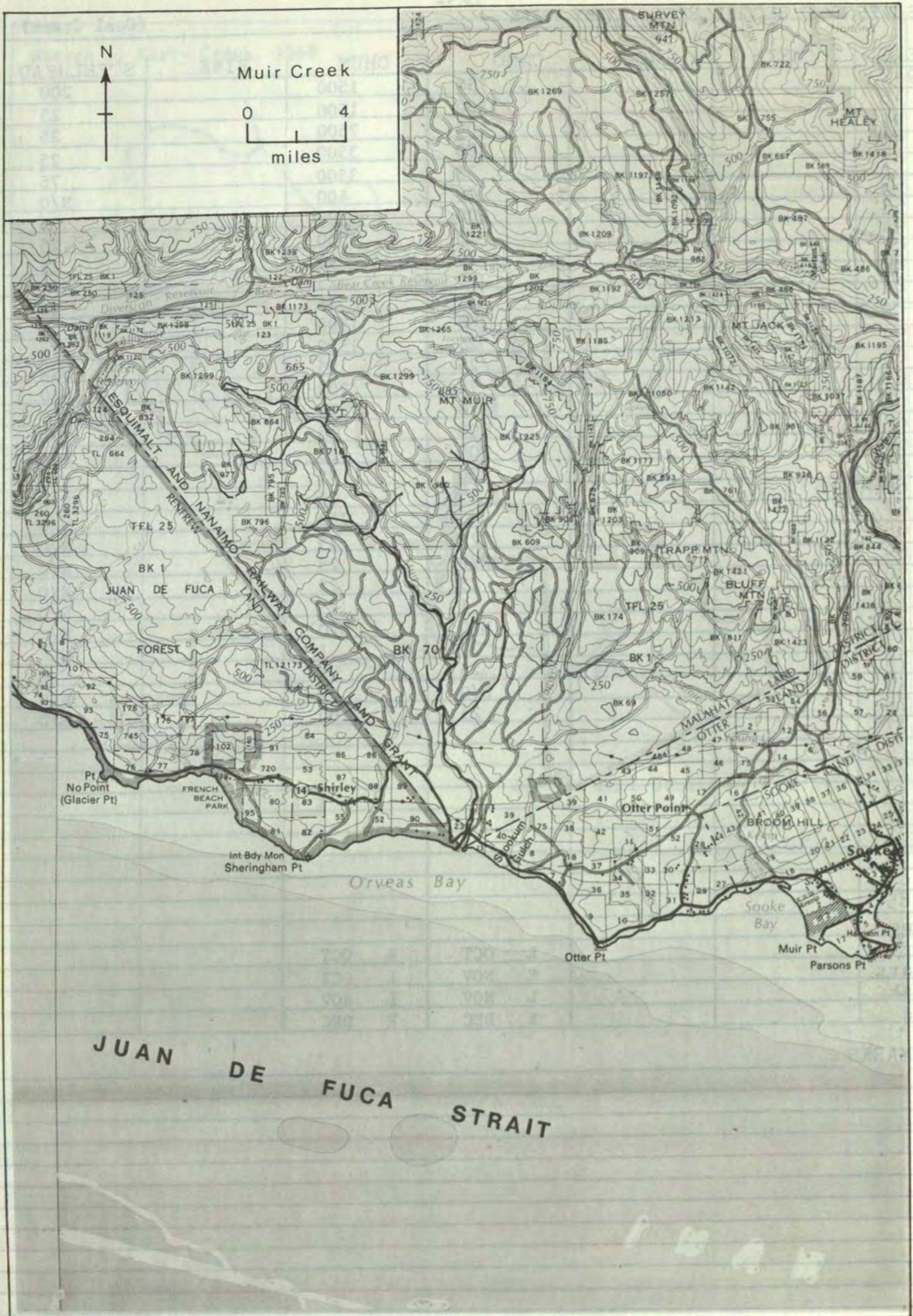
ESCAPEMENT RECORD FOR KIRBY CREEK

42

(Coal Creek)

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947			750	1500		200
48			1500	1500		25
49			400	7500		25
50			200	3500		25
51			75	3500		75
52			75	400		N/O
53			200	1500		25
54			25	1500		25
55			25	25		25
56			25	25		N/O
57			25	75		25
58			200	1500		25
59			25	1500		N/O
60			25	25		UNK
61			75	200		N/O
62			N/O	75		N/O
63			200	25		N/O
64			200	25		N/O
65			200	75		N/O
66			25	25		N/O
67			25	750		N/O
68			25	25		N/O
69			N/O	25		N/O
70			25	25		N/O
71			1	25		N/O
72			N/O	N/O		N/O
73			23	1		UNK
74			25	25		N/O
75			N/O	25		N/O
76			25	200		N/O
77						
78						
79						
80						
81						
82						
83						
84						
85						
Time						
Arr.			L OCT	L OCT		
Start			E NOV	L OCT		
Peak			L NOV	L NOV		
End			E DEC	E DEC		

REMARKS



NAME OF STREAM MUIR CREEK

44

CONSERVATION DISTRICT 3STATISTICAL AREA 20LOCATION OF MOUTH Flows S. into Orveas Bay, E. of Sheringham Pt., Malahat Dist.LENGTH 3 MI. POSITION 48 123 SW.WIDTH FT. DRAINAGE SQ. MI.COMPOSITION: BEDROCK BOULDER COARSE FINE SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

0.0 - 2.5

2.5 - 5.0

5.0 - 7.5

7.5 - 10.0

> 10.0

WETTED AREA SQ. YD.SPAWNING AREA SQ. YD.DISCHARGE CFS MAX MINTEMPERATURE BARRIERS OR POINTS OF DIFFICULT ASCENT - Impassable falls at 5 mi.

SPAWNING DISTRIBUTION:

SPECIES

SECTION OF STREAM USED

SCKEYE

CHINOOK

COHO

lower 3 mi.

CHUM

lower 3 mi.

PINK (ODD YR)

PINK (EVEN YR)

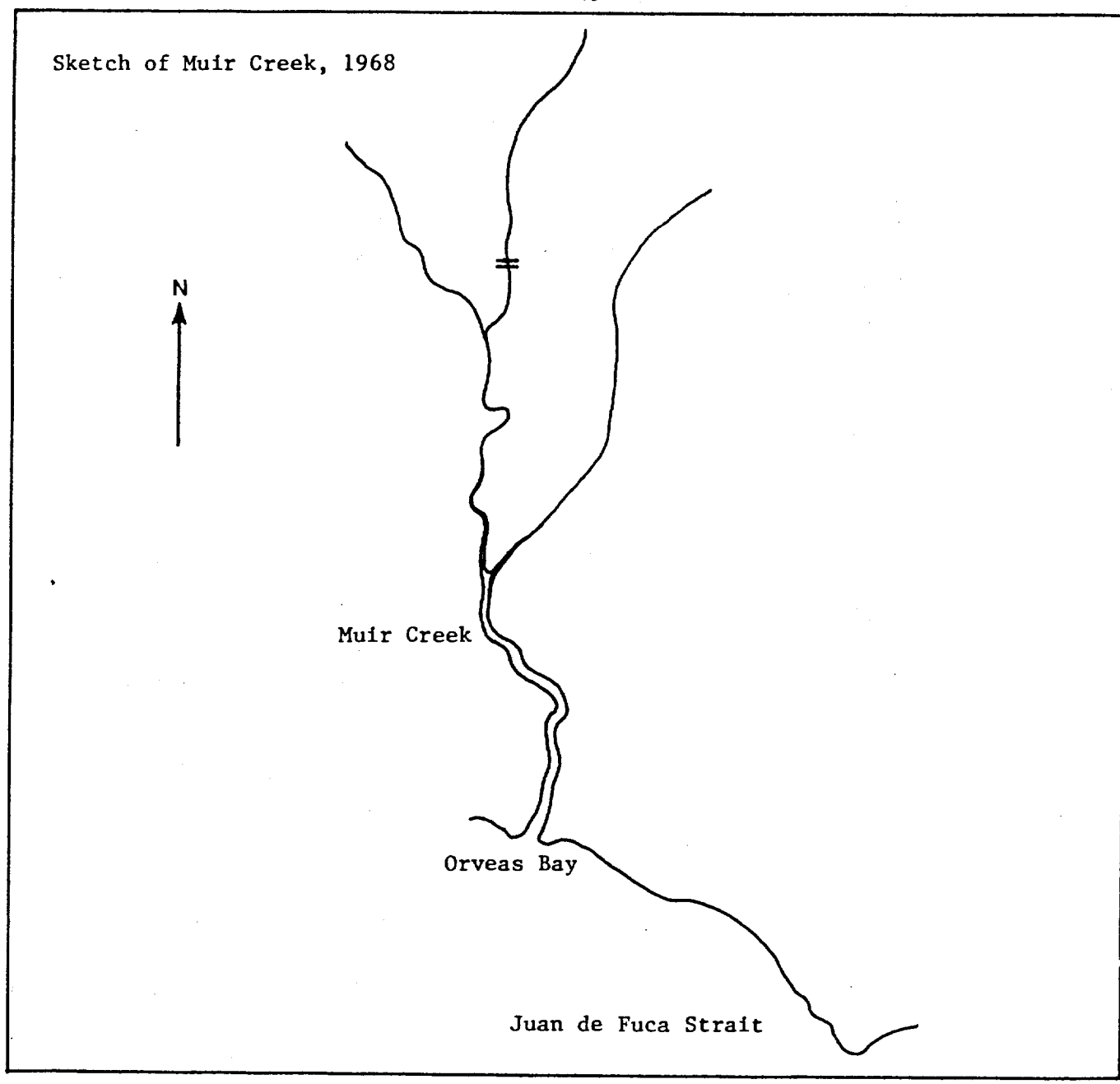
STEELHEAD

POTENTIAL OF INACCESSIBLE PORTION OF STREAM

GENERAL REMARKS:

- This stream has a silting problem due to previous logging. (1949)
- Escapements have been reduced by a heavy net fishery in Juan de Fuca Strait and by repeated spraying of road-side vegetation. (1967)
- Quick runoff and silting are a continuing problem caused from previous logging. (1972)

Sketch of Muir Creek, 1968



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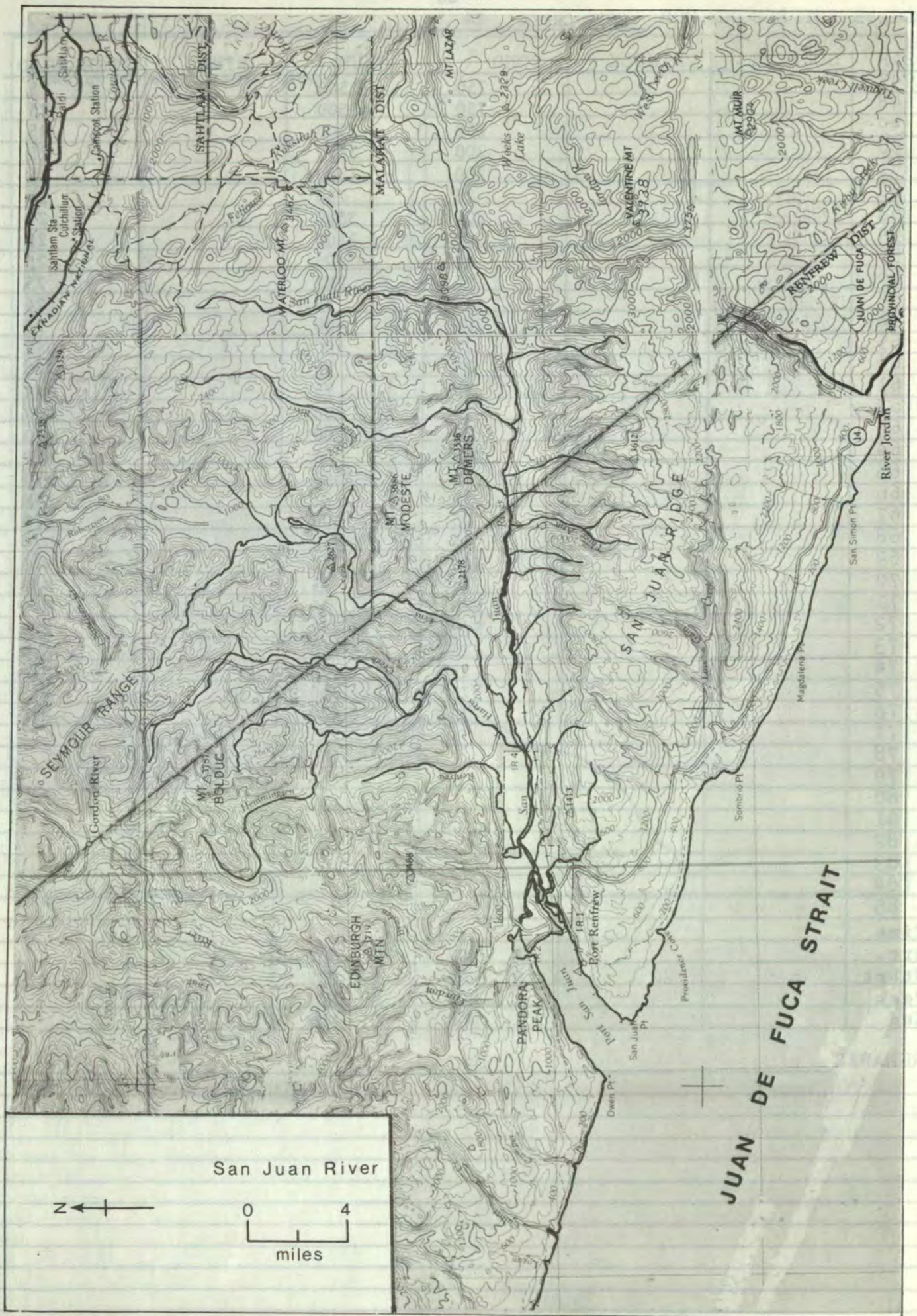
R

ESCAPEMENT RECORD FOR MUIR CREEK

46

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947						
48			750	1500		75
49			750	3500		75
50			750	1500		25
51			200	3500		25
52			200	1500		25
53			25	750		25
54			200	3500		25
55			200	3500		25
56			25	400		25
57			25	UNK		25
58			25	25		25
59			200	400		25
60			25	400		UNK
61			25	25		N/O
62			25	75		UNK
63			25	200		UNK
64			25	200		N/O
65			200	750		N/O
66			75	3500		N/O
67			25	25		UNK
68			25	400		N/O
69			25	750		N/O
70			N/O	750		N/O
71			25	25		N/O
72			N/O	75		N/O
73			N/O	200		UNK
74			25	750		N/O
75			25	400		N/O
76			75	200		N/O
77			75	400		N/O
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Time						
Arr.						
Start			E NOV	E NOV		
Peak			M NOV	E NOV		
End			L NOV	M NOV		
			L NOV	L NOV		

REMARKS



NAME OF STREAM SAN JUAN RIVERCONSERVATION DISTRICT 3 STATISTICAL AREA 20LOCATION OF MOUTH Flows S. and W. into Port San Juan, Renfrew Dist.POSITION 48 124 NE.LENGTH MI. WIDTH FT. DRAINAGE 254 SQ. MI.COMPOSITION: BEDROCK BOULDER COARSE FINE SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

0.0 - 2.52.5 - 5.05.0 - 7.57.5 - 10.0> 10.0WETTED AREA SQ. YD. SPAWNING AREA SQ. YD.DISCHARGE 1854 CFS MAX 38,000 cfs 15/01/61 MIN 38.5 cfs 29/08/61TEMPERATURE BARRIERS OR POINTS OF DIFFICULT ASCENT - Passable falls (Hemingsen Falls) at 12 mi.

SPAWNING DISTRIBUTION:

SPECIES

SECTION OF STREAM USED

SCKEYE

CHINOOK

in mainstream

COHO

throughout tributaries

CHUM

PINK (ODD YR)

PINK (EVEN YR)

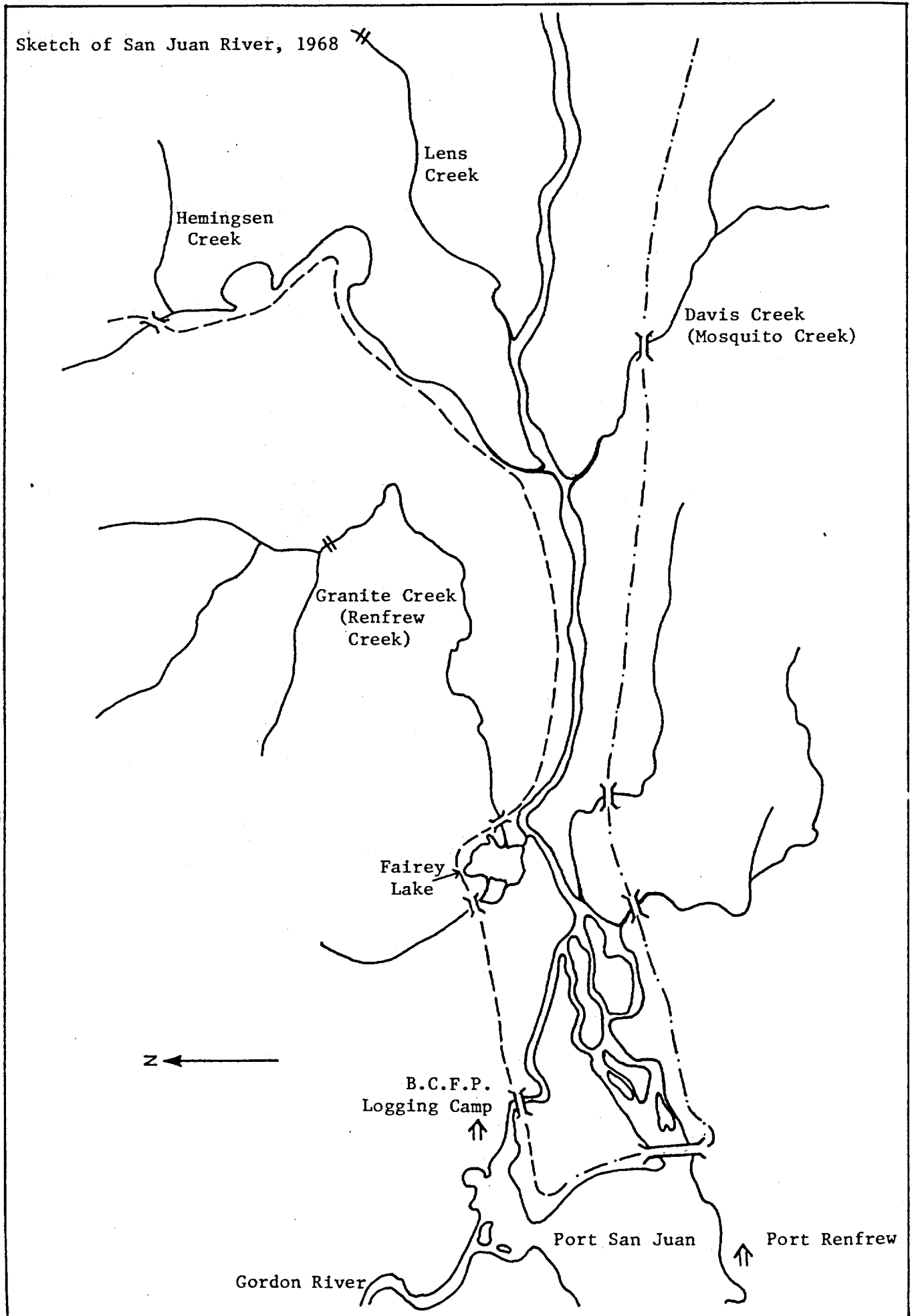
STEELHEAD

POTENTIAL OF INACCESSIBLE PORTION OF STREAM

GENERAL REMARKS:

- Severe floods scoured spawning beds heavily. (1960)
- Severe flood damage occurred in the lower reaches and to some extent upstream. (1962)
- Various log jams on Mosquito Creek should be cleared. (1973)
- Some stream clearing work was done this year. (1975)

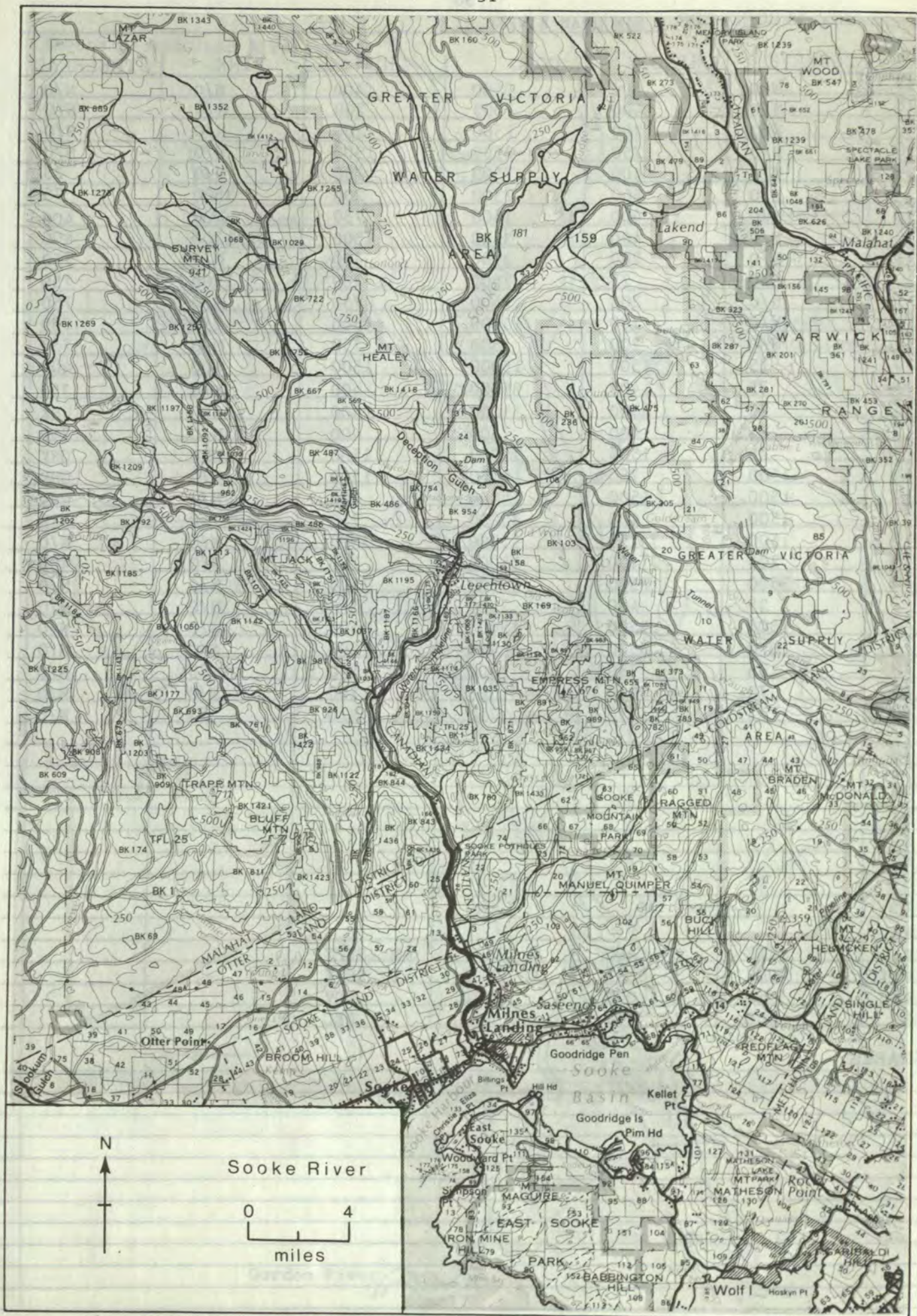
Sketch of San Juan River, 1968



ESCAPEMENT RECORD FOR SAN JUAN RIVER

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947	N/O	400	3500	400	N/O	N/O
48	N/O	400	7000	200	N/O	N/O
49	N/O	200	1500	200	N/O	N/O
50	N/O	750	3500	1500	N/O	N/O
51	N/O	400	7500	750	N/O	25
52	N/O	400	7500	200	3500	75
53	N/O	400	1500	750	N/O	400
54	N/O	400	3500	1500	1500	400
55	N/O	200	3500	400	N/O	750
56	N/O	25	7500	200	1500	750
57	N/O	200	7500	750	N/O	750
58	N/O	200	15000	1500	750	750
59	N/O	200	7500	750	N/O	750
60	75	1500	15000	1500	400	400
61	75	400	35000	400	N/O	750
62	75	400	15000	1500	400	UNK
63	25	400	7500	400	N/O	750
64	25	1500	75000	1500	UNK	UNK
65	1500	750	15000	1500	N/O	UNK
66	1500	1500	15000	400	N/O	UNK
67	1500	400	35000	750	750	UNK
68	1500	750	15000	3500	N/O	N/O
69	200	400	15000	200	N/O	400
70	3500	7500	75000	7500	N/O	N/O
71	25	400	7500	N/O	N/O	N/O
72	1500	3500	15000	3500	N/O	N/O
73	N/O	3500	15000	200	N/O	N/O
74	75	750	7500	N/O	75	N/O
75	75	200	1500	200	N/O	N/O
76	75	75	750	N/O	75	N/O
77						
78						
79						
80						
81						
82						
83						
84						
85						
Time						
Arr.	JUNE	M AUG	M SEPT	M SEPT		
Start	JUNE	L SEPT	L SEPT	L SEPT		
Peak	JUNE	E OCT	OCT	NOV		
End	OCT	L NOV	L NOV	DEC		

REMARKS



NAME OF STREAM SOOKE RIVER
 CONSERVATION DISTRICT 3 STATISTICAL AREA 20
 LOCATION OF MOUTH Flows S. into Sooke Hr., Sooke Dist.
 POSITION 48 123 SW.
 LENGTH 4 MI. WIDTH FT. DRAINAGE SQ. MI.
 COMPOSITION: BEDROCK BOULDER COARSE FINE
 SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

0.0 - 2.5
2.5 - 5.0
5.0 - 7.5
7.5 - 10.0
> 10.0

WETTED AREA SQ. YD. SPAWNING AREA SQ. YD.DISCHARGE CFS MAX MIN TEMPERATURE BARRIERS OR POINTS OF DIFFICULT ASCENT

- Impassable falls at 4 mi.

SPAWNING DISTRIBUTION:

SPECIES	SECTION OF STREAM USED
SOCKEYE	
CHINOOK	throughout
COHO	
CHUM	throughout
PINK (ODD YR)	
PINK (EVEN YR)	
STEELHEAD	

POTENTIAL OF INACCESSIBLE PORTION OF STREAM

- There are 25 mi. of valuable coho spawning and rearing areas beyond Sooke Falls in the main Leech and West Leech River.

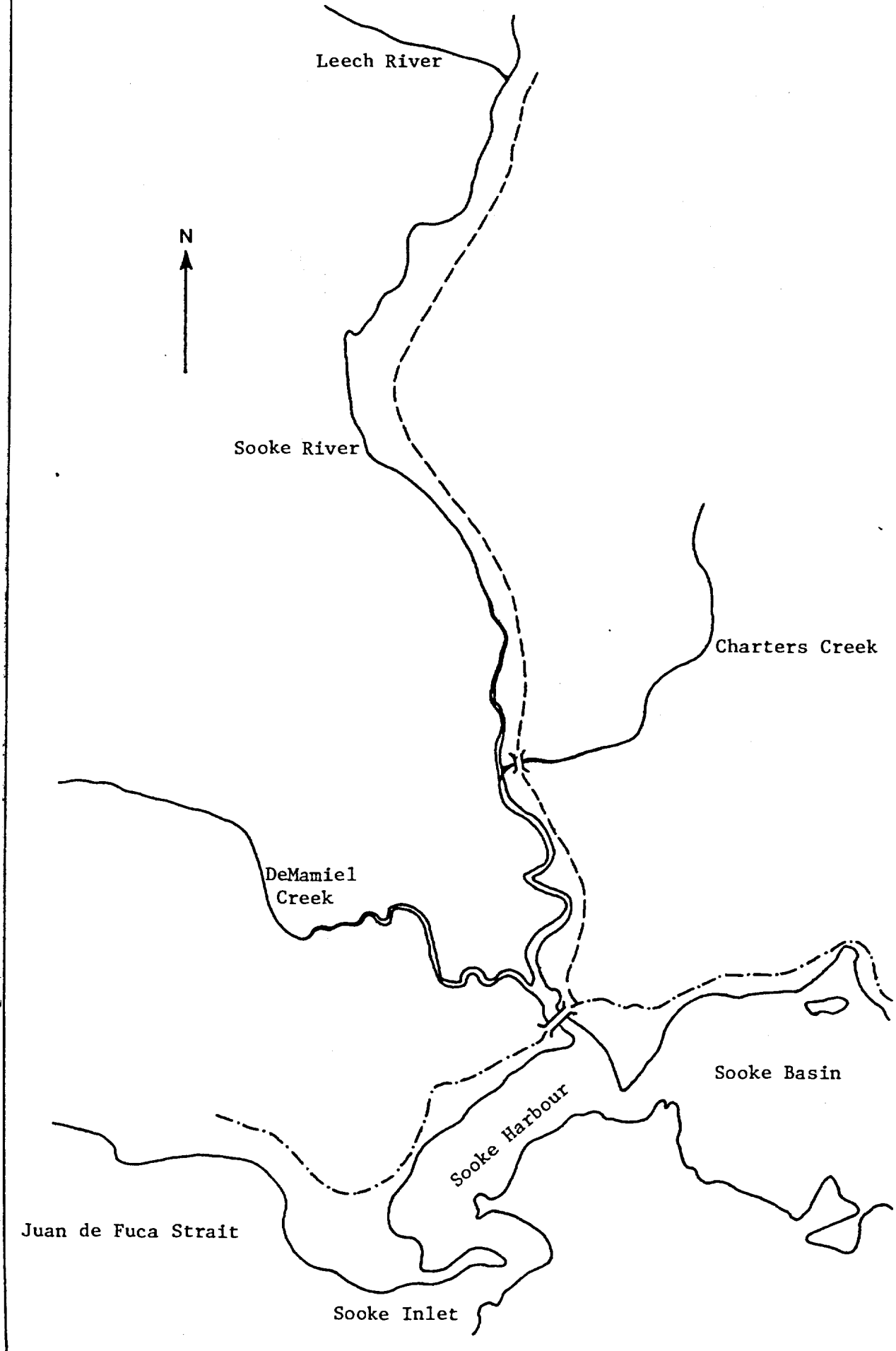
GENERAL REMARKS:

- Flood conditions resulted in scouring of spawning beds and great losses of spawn. (1961)

- Sooke River is dammed at Sooke Lake by the Victoria Water Board. (1967)

- The storage dam at Sooke Lake has been raised to 23.5 ft. Building the dam, clearing the reservoir and relocating the roads were completed in 1970.

Sketch of Sooke River, 1968



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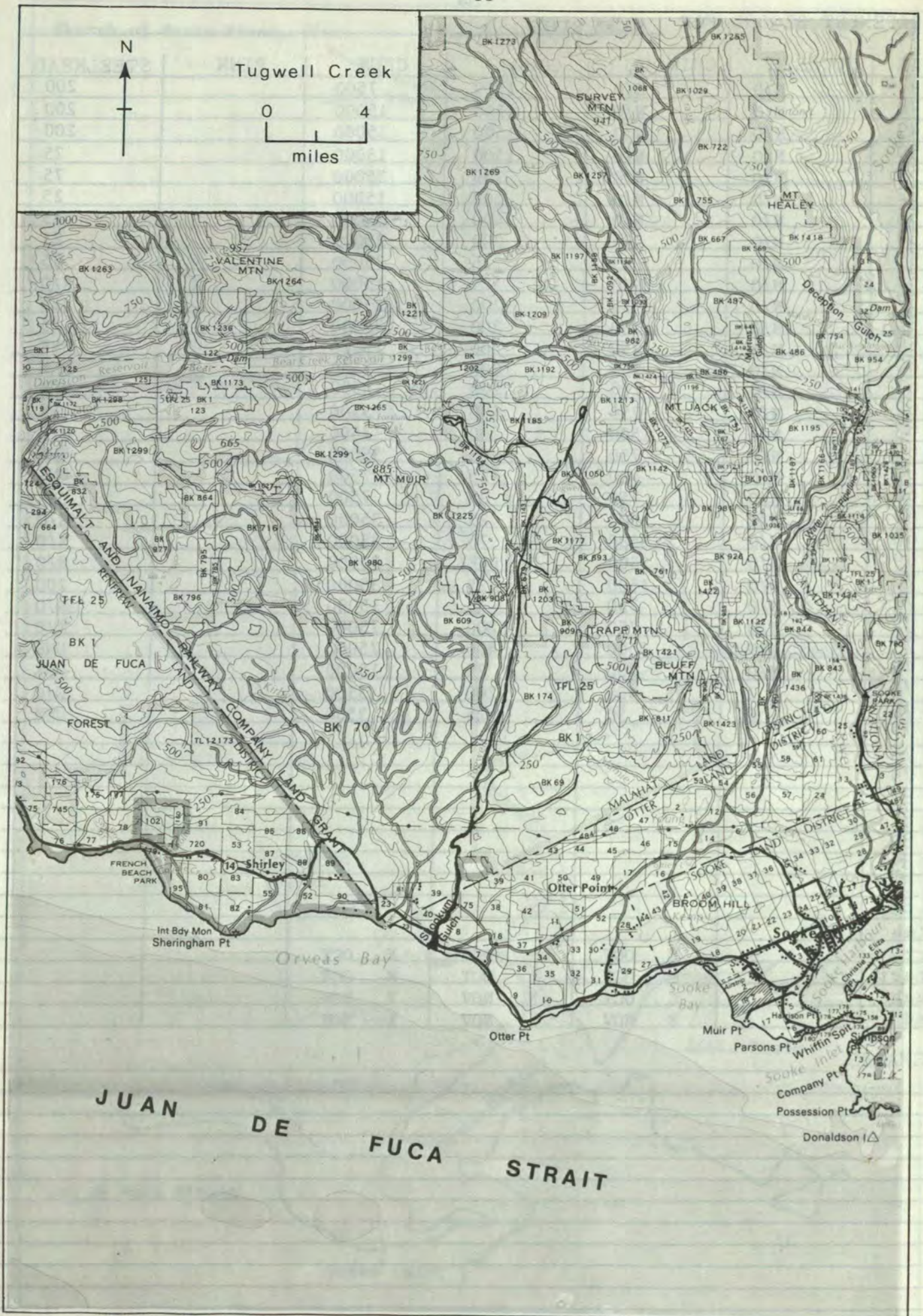
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ESCAPEMENT RECORD FOR SOOKE RIVER

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947	N/O	N/O	1500	7500		200
48	N/O	N/O	3500	15000		200
49	N/O	N/O	1500	35000		200
50	N/O	N/O	1500	15000		25
51	N/O	N/O	750	35000		75
52	N/O	N/O	200	15000		25
53	N/O	N/O	400	35000		25
54	N/O	75	200	35000		25
55	N/O	25	75	35000		25
56	N/O	N/O	25	15000		N/O
57	N/O	25	200	15000		25
58	N/O	200	200	35000		25
59	N/O	250	200	15000		UNK
60	N/O	350	200	3500		UNK
61	N/O	200	200	3500		UNK
62	N/O	750	400	3500		UNK
63	N/O	1500	200	7500		N/O
64	N/O	750	75	35000		UNK
65	N/O	400	25	7500		UNK
66	N/O	750	25	7500		UNK
67	N/O	1500	25	35000		N/O
68	N/O	3500	75	35000		N/O
69	N/O	1500	UNK	35000		N/O
70	N/O	1500	25	35000		200
71	N/O	750	75	15000		N/O
72	N/O	750	200	75000		N/O
73	UNK	3500	400	75000		UNK
74	N/O	200	75	15000		UNK
75	2	400	200	3500		N/O
76	8	30	25	3500		UNK
77						
78						
79						
80						
81						
82						
83						
84						
85						
Time						
Arr.		E OCT	M OCT	E OCT		
Start		M OCT	L OCT	M OCT		
Peak		L OCT	M NOV	E NOV		
End		E NOV	L NOV	L NOV		

REMARKS



NAME OF STREAM TUGWELL CREEKCONSERVATION DISTRICT 3 STATISTICAL AREA 20LOCATION OF MOUTH Flows S. into Orveas Bay, NW. of Otter Pt., Otter Dist.POSITION 48 123 SW.LENGTH 3 MI. WIDTH FT. DRAINAGE SQ. MI.COMPOSITION: BEDROCK BOULDER COARSE FINE SILT & SAND UNCLASSIFIED

GRADIENT:

FALL IN FT/000

0.0 - 2.52.5 - 5.05.0 - 7.57.5 - 10.0> 10.0WETTED AREA SQ. YD.SPAWNING AREA SQ. YD.DISCHARGE CFS MAX MIN TEMPERATURE BARRIERS OR POINTS OF DIFFICULT ASCENT - Impassable falls at 5 mi.

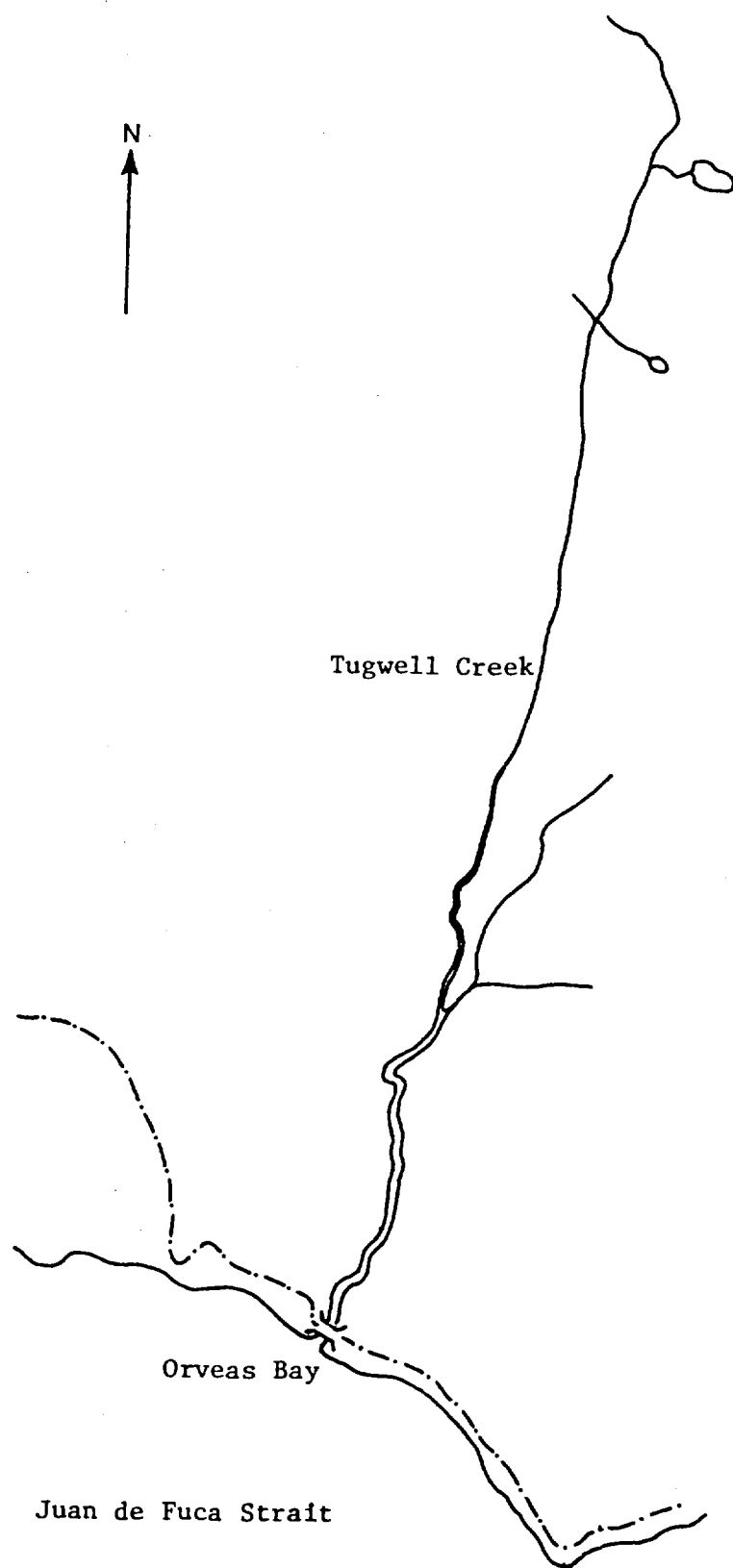
SPAWNING DISTRIBUTION:

SPECIES SECTION OF STREAM USED SCKEYE CHINOOK COHO to 3 mi.CHUM lower 500 yds.PINK (ODD YR) PINK (EVEN YR) STEELHEAD POTENTIAL OF INACCESSIBLE PORTION OF STREAM

GENERAL REMARKS:

- The escapement of this stream is affected by heavy commercial fishing during migration, logging and chemical spraying of roadside vegetation. (1967)

Sketch of Tugwell Creek, 1968



ESCAPEMENT RECORD FOR TUGWELL CREEK

YEAR	SOCKEYE	CHINOOK	COHO	CHUM	PINK	STEELHEAD
1947						
48			400	1500		200
49			750	750		75
50			400	1500		25
51			200	3500		25
52			200	1500		75
53			25	25		25
54			200	3500		25
55			200	1500		25
56			25	750		25
57			25	25		N/O
58			25	25		25
59			200	750		25
60			25	3500		UNK
61			25	25		N/O
62			25	25		N/O
63			75	750		N/O
64			25	25		N/O
65			25	200		N/O
66			25	25		UNK
67			25	25		UNK
68			25	75		N/O
69			25	75		N/O
70			N/O	25		UNK
71			25	25		N/O
72			N/O	25		N/O
73			25	25		N/O
74			35	11		UNK
75			200	N/O		N/O
76			75	25		N/O
77			N/O	25		N/O
78						
79						
80						
81						
82						
83						
84						
85						
Time						
Arr.			E NOV	L OCT		
Start			E NOV	M NOV		
Peak			M NOV	L NOV		
End			L NOV	L NOV		

REMARKS

METRIC EQUIVALENTSLength

cm.	=	.3937	in.
meter	=	3.28	ft.
meter	=	1.094	yd.
kilometer	=	.621	mi.
inch	=	2.54	cm.
foot	=	.3048	m.
yard	=	.9144	m.
mile	=	1.61	km.

Area

sq. cm.	=	.1550	sq. in.
sq. m.	=	10.76	sq. ft.
sq. m.	=	1.196	sq. yd.
sq. km.	=	.386	sq. mi.
sq. in.	=	6.45	sq. cm.
sq. ft.	=	.0929	sq. m.
sq. yd.	=	.836	sq. m.
sq. mi.	=	2.59	sq. km.
acre	=	.405	ha.
hectare	=	2.47	acres
acre	=	43560	sq. ft.

Volume

cu. cm.	=	.061	cu. in.
cu. m.	=	35.315	cu. ft.
cu. m.	=	1.308	cu. yd.
cu. in.	=	16.38	cu. cm.
cu. ft.	=	.028	cu. m.
cu. yd.	=	.7645	cu. m.

Capacity

liter	=	.0353	cu. ft.
liter	=	.21998	gal. (Br.)
liter	=	61.023	cu. in.
cu. in.	=	.0164	l.
cu. ft.	=	28.32	l.
gallon	=	4.5459	l. (Br.)

Weight

gram	=	15.432	grs.
gram	=	.0353	oz.
kilogram	=	2.2046	lbs.
kilogram	=	.0011	ton (sht)
met. ton	=	1.1025	ton (sht)
grain	=	.0648	g.

ounce	=	28.35	g.
pound	=	.454	kg.
ton (sht)	=	907.18	kg.
ton (sht)	=	.907	met. ton
ton (sht)	=	2000	lbs.

Degrees Centigrade = $5/9$ (Degrees Fahr. - 32)
 Degrees Fahrenheit = $9/5$ (Degrees Cent.) + 32

WATER QUANTITIES AND FLOW MEASUREMENT

1 cubic foot per second (cfs) or second foot	=	373.2 gallons per min. (gpm)
1 cubic foot per second (cfs) or second foot	=	.537408 million gallons
1 second foot	=	approx. 2 acre-feet per day
1 second foot	=	86400 cubic feet per day
1 million gallons per day	=	1.86 cfs
1 acre-foot	=	43560 cubic feet or 271379 gal.
1 cubic foot of water	=	6.23 gal. and weighs 62.4 lbs.
1 cubic meter per second	=	35.31 cubic feet per second (cfs)
1 meter per second	=	3.28 feet per second
1233.5 cubic meters	=	1 acre-foot

